Software Analysis: Dynamic Views of the LavaRápido System

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Introduction

This document shows how LavaRápido behaves when someone uses it: what happens, in which order, and between which screens and services. It is organized in four parts:

1. **Use cases by actor**, with their «`include`»/«`extend`» relationships.
2. **Sequence diagrams** of the main flows: from the screen to the gateway, from the gateway to the service, and from the service to the database or to another service.
3. **Activity diagrams** of the business processes.
4. **State diagrams** of the entities that change status (booking, payment, service execution, promotion and inspection report).

The diagrams are written in Mermaid. They were checked against the code and the running system on October 9, 2026, and updated against the code on October 15, 2026 with the account lockout, automatic assignment, operator choice, the service start window, the alert for bookings without an operator and the inspection report. The screen inventory, with screenshots by role, is at the end.

The full specification of each requirement and use case (actors, precondition, main flow, alternative flows and postcondition) is in the *Software Requirements Specification*, sections 5 and 7.3; here they are cited by their identifier (RF-XXX, CUXX).

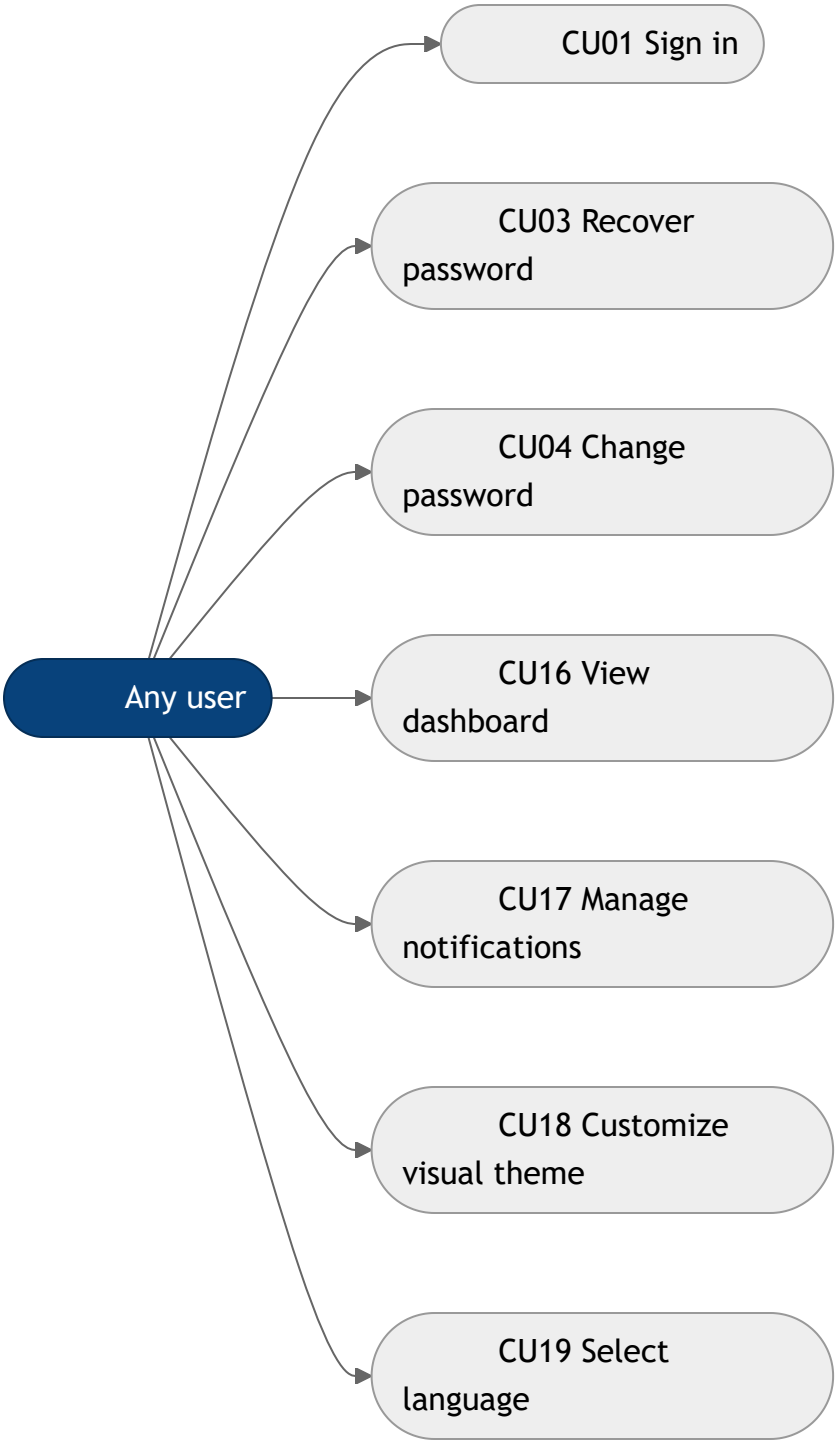
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1. Use Cases by Actor

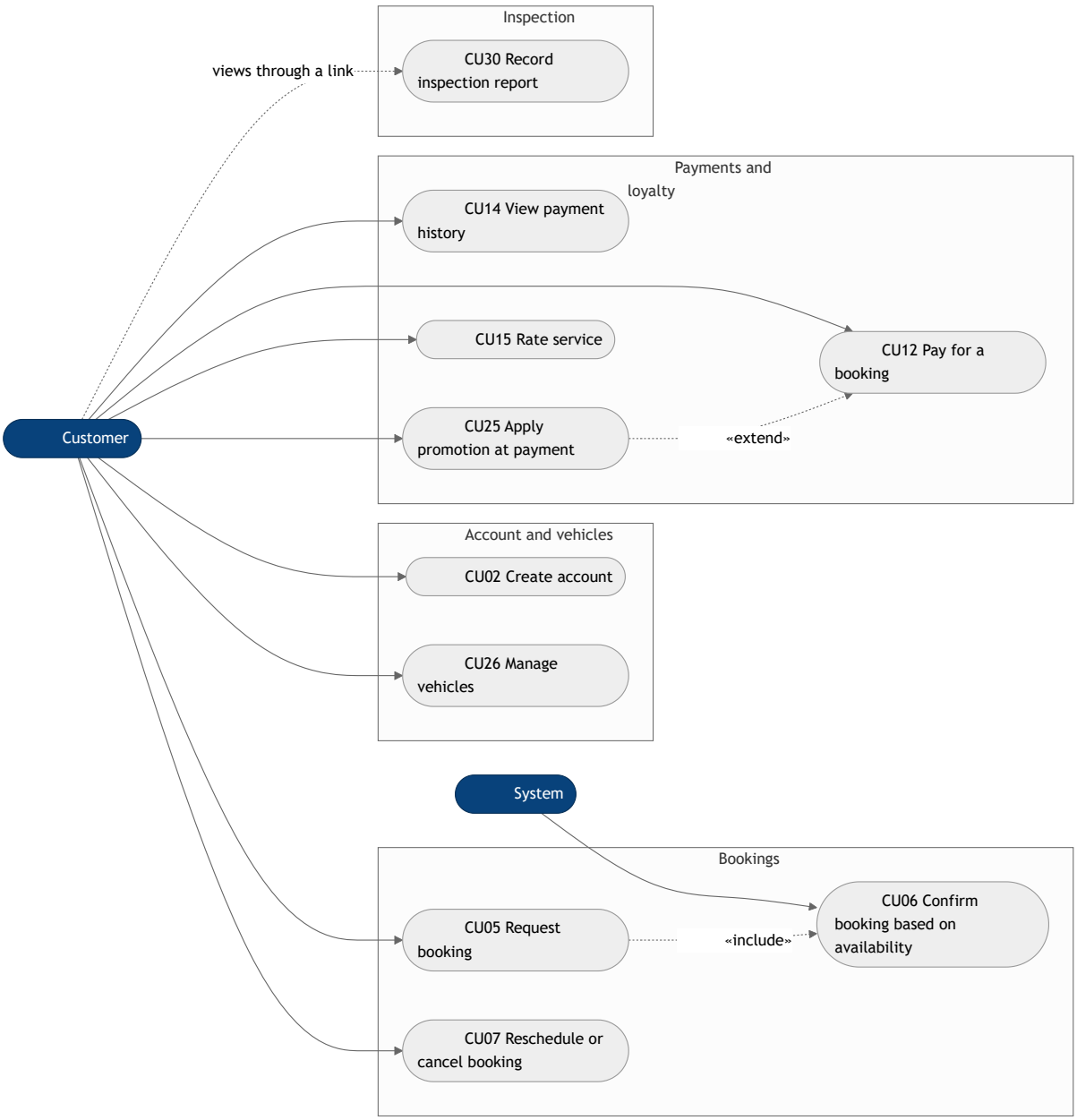
1.1 Use Case Diagram

The 30 use cases are presented in four diagrams, one per actor, grouped by module.

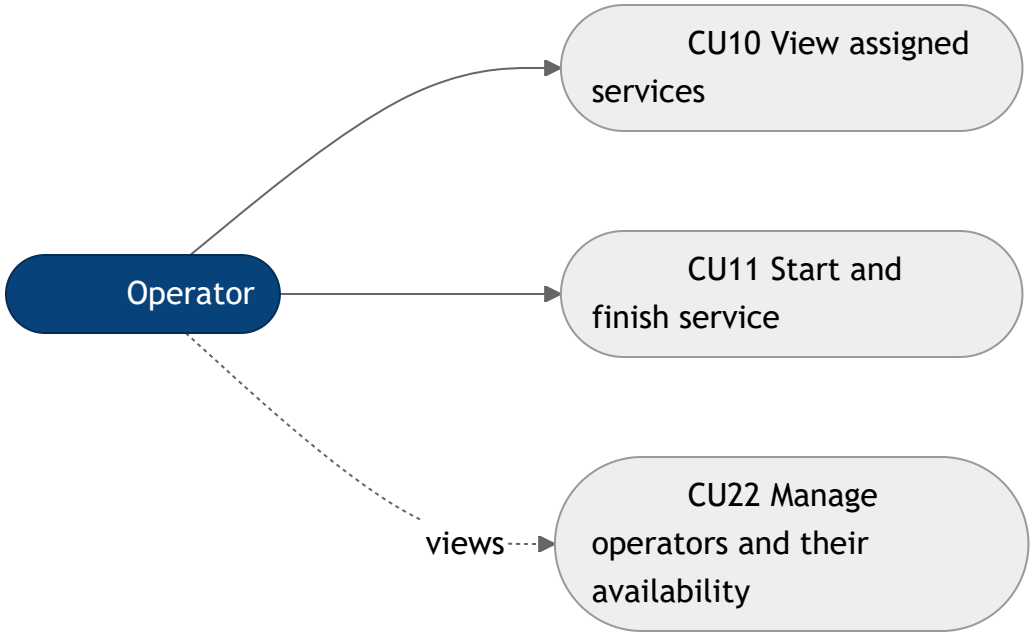
Use cases common to the three roles



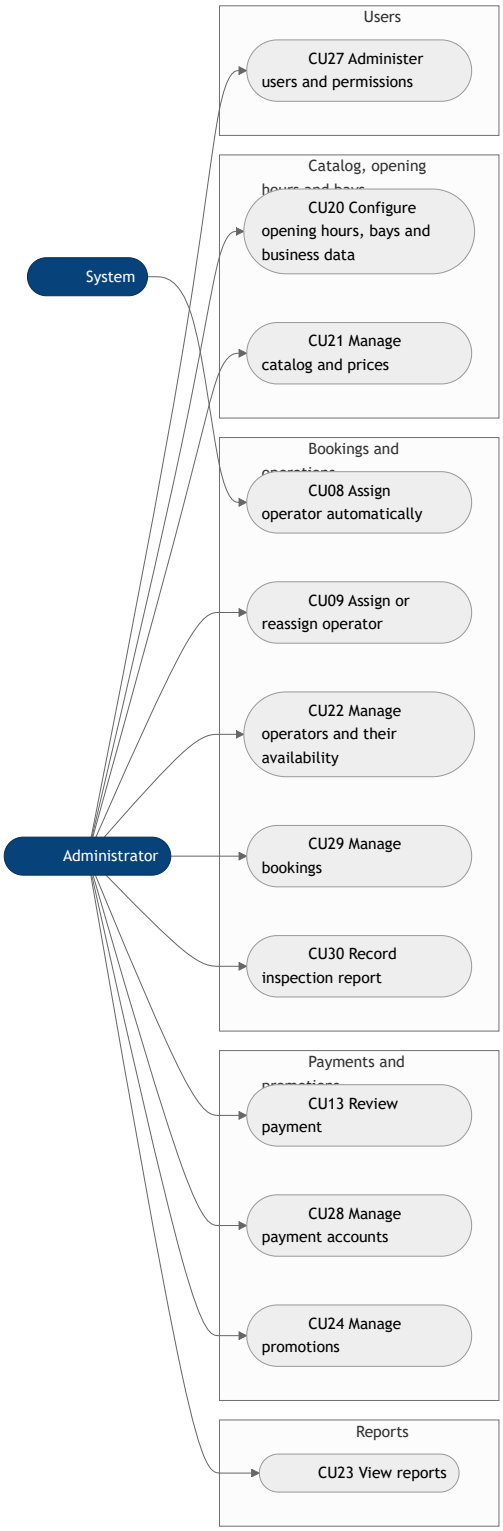
Customer use cases



Operator use cases



Administrator and System use cases



Note. «include» means that CU05 always runs CU06; «extend» means that CU25 optionally extends CU12. CU22 appears as view-only for the operator and as management for the

administrator, who is the one recording shifts, absences and the operator profile (RF-022, RF-038). The system runs CU08 each time a booking is confirmed. The customer does not record the inspection report (CU30): they view it through the link the administrator shares.

1.2 Actor and Use Case Matrix

Actor	Use cases
All roles	CU01, CU03, CU04, CU16, CU17, CU18, CU19
Customer	CU02, CU05, CU06, CU07, CU12, CU14, CU15, CU25, CU26, CU30 (view only)
Operator	CU10, CU11, CU22 (view only)
Administrator	CU09, CU13, CU20, CU21, CU22, CU23, CU24, CU27, CU28, CU29, CU30
System	CU06 (included in CU05), CU08

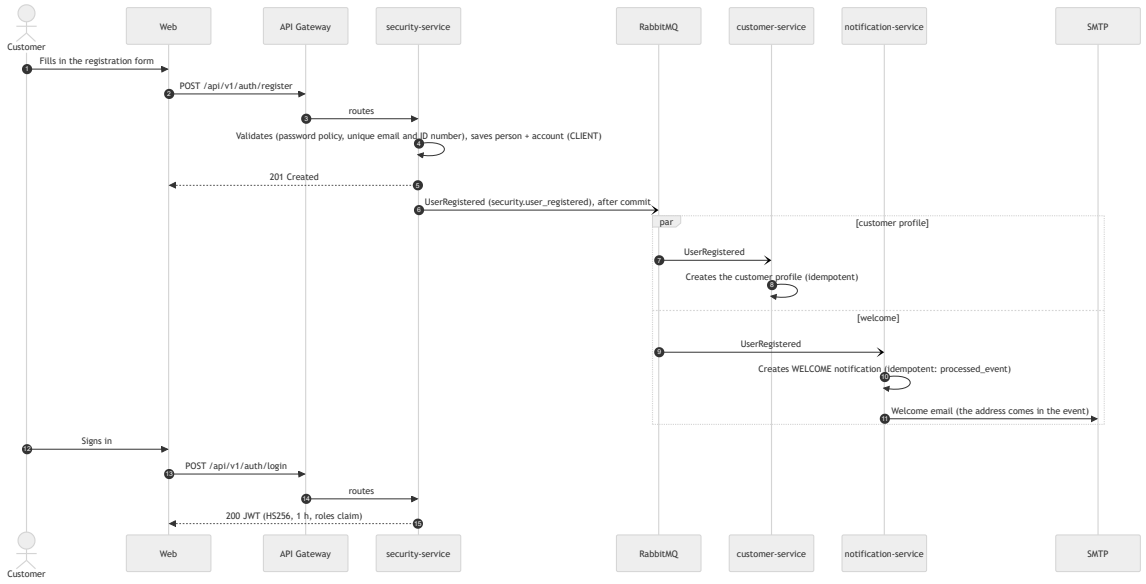
The specification of each of the 30 use cases (precondition, main flow, alternative flows, postcondition, status) is in the *Software Requirements Specification*, Tables 23 to 52.

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2. Sequence Diagrams

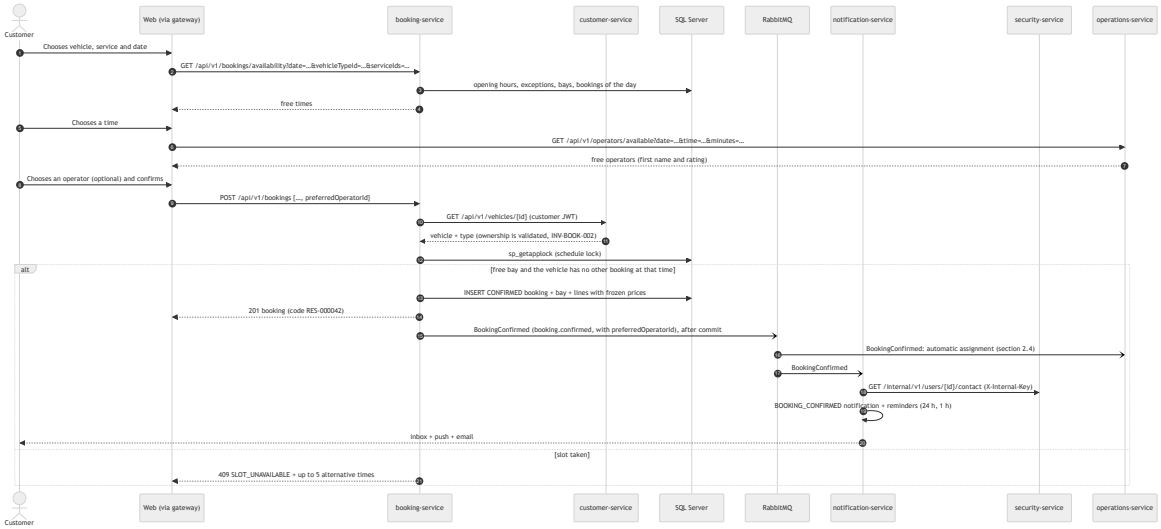
Each diagram follows the path of the request (screen, gateway, services, database and RabbitMQ) with the class and endpoint names from the code. The first five are based on the diagrams in `vehicle-w-docs/08-uml/diagrams/source/`; the next ones cover loyalty, roles and permissions, the account lockout and the inspection report.

2.1 Registration and Welcome (RF-001, RF-018)



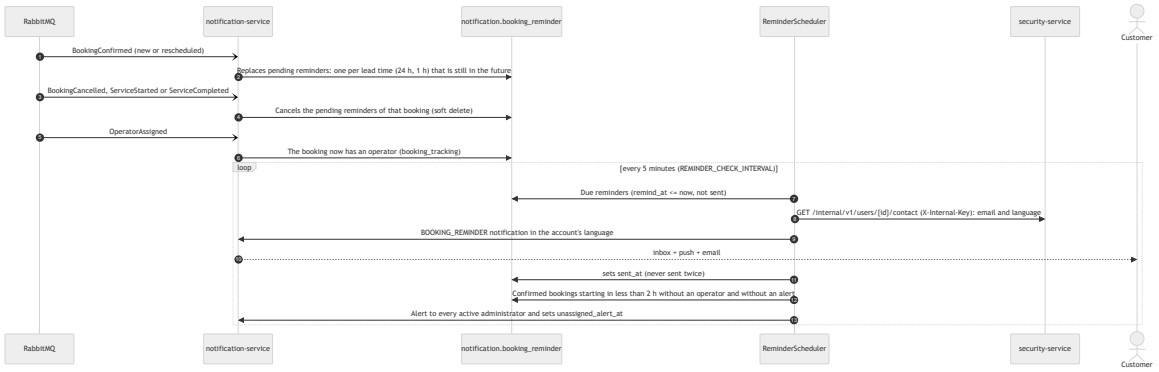
Source: `security-service` (`AccountRegistrationService`, `AuthenticationService`), event `security.user_registered`.

2.2 Booking a Wash (RF-005, RF-006, RF-010)



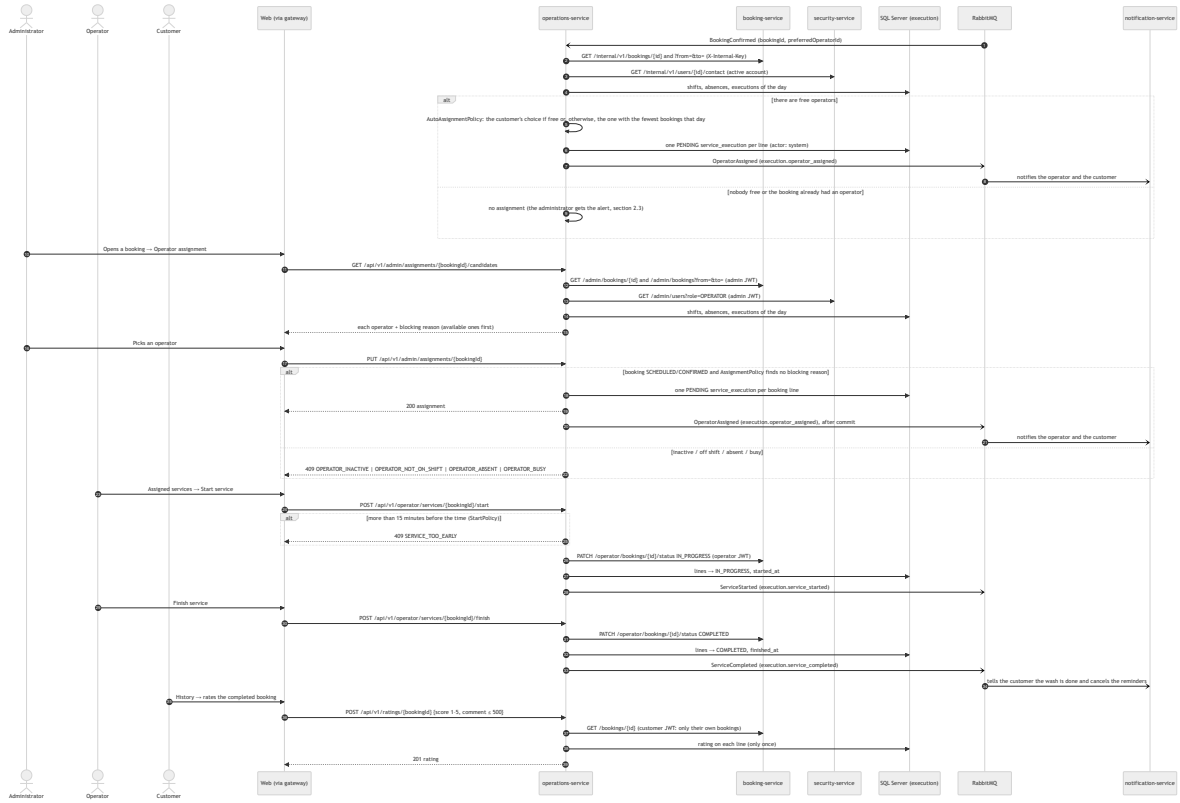
Source: `booking-service` (`BookingService.confirmNew`), `operations-service` (`AvailableOperatorsController`), contracts in `07-api/contracts/openapi/`.

2.3 Reminders and Alert for Bookings Without an Operator (ADR-011)



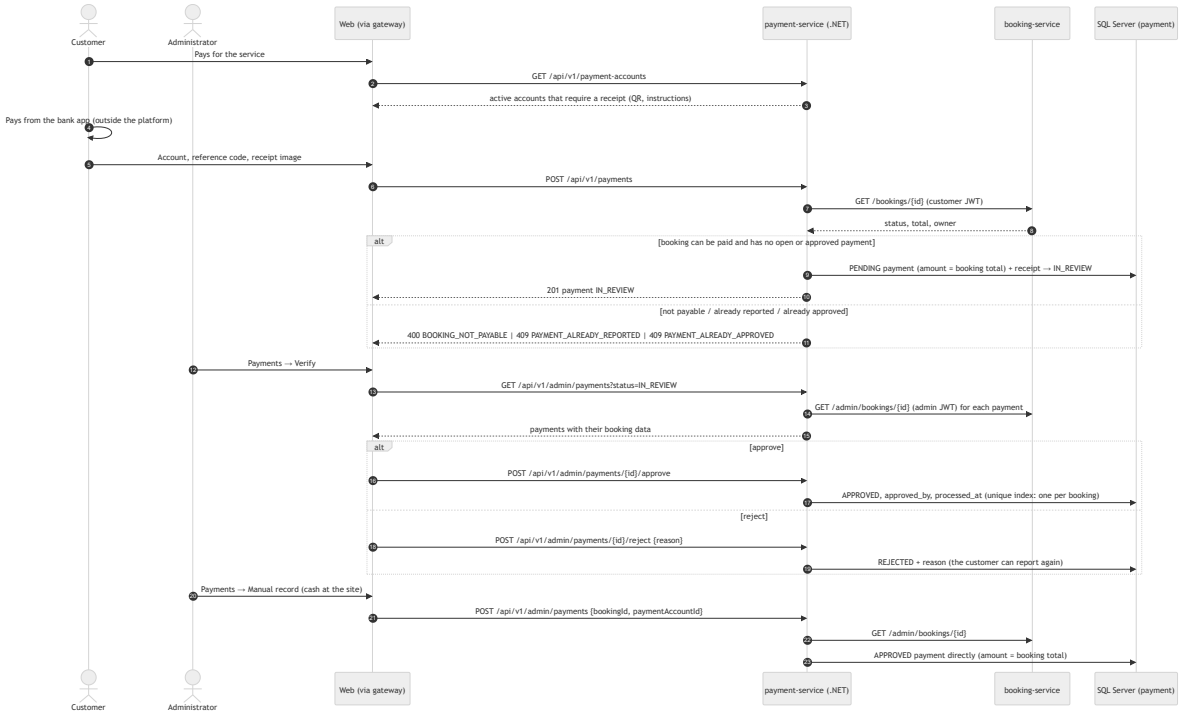
Source: `notification-service` (`BookingReminderService`, `ReminderScheduler`, `StaffNotificationService`, migration 031).

2.4 Operator Assignment, Execution and Rating (RF-008, RF-009, RF-010, RF-011, RF-012, RF-012b)



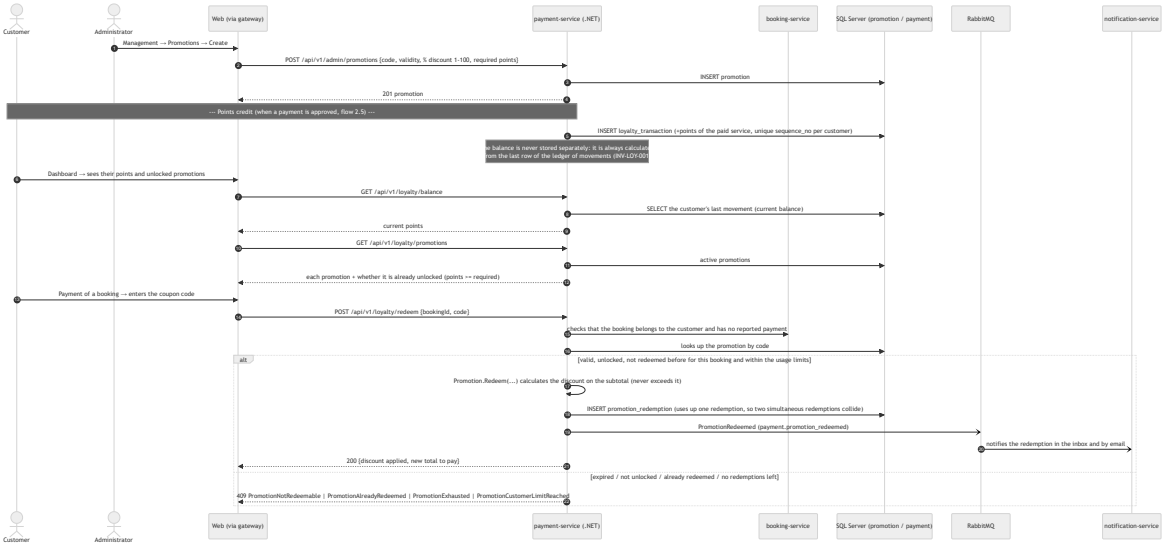
Source: `operations-service` (`BookingConfirmedListener`, `OperationsService.autoAssign / candidates / assign / start / finish / rate`, `AssignmentPolicy`, `AutoAssignmentPolicy`, `StartPolicy`, ADR-014).

2.5 Payment by Receipt and Review (RF-013, RF-013b, RF-014)



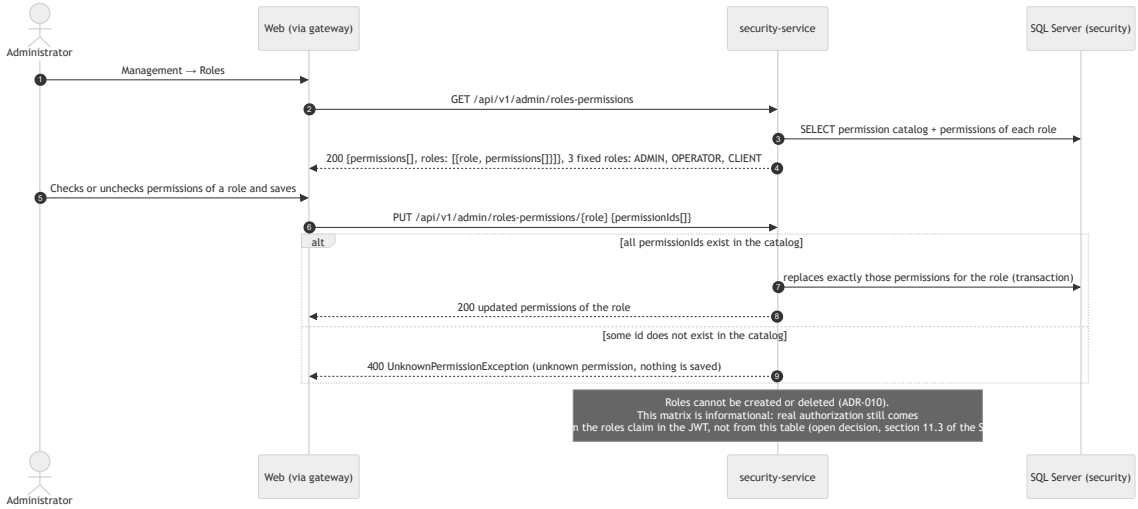
Source: `payment-service` (`PaymentApplicationService.ReportPaymentAsync / ApproveAsync / RejectAsync / RegisterManualAsync`, `Payment` aggregate, ADR-008, ADR-013).

2.6 Coupon Redemption and Loyalty Points (RF-024, RF-025, RF-026)



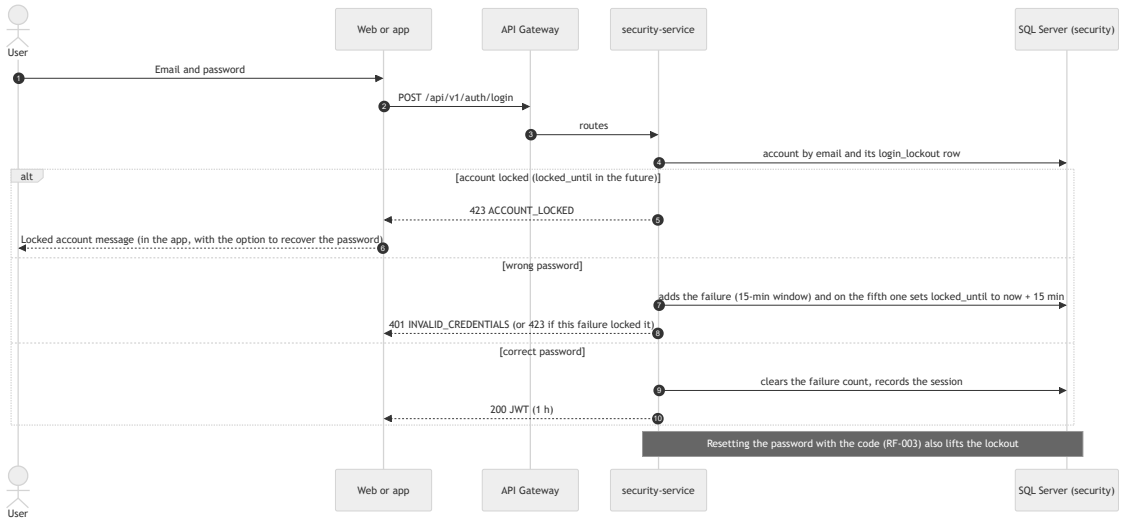
Source: `payment-service` (`LoyaltyController`, `CustomerPromotionService.RedeemAsync`, `Promotion.Redeem` aggregate, `LoyaltyLedgerRepository`), ADR-015.

2.7 Roles and Permissions Management (RF-029)



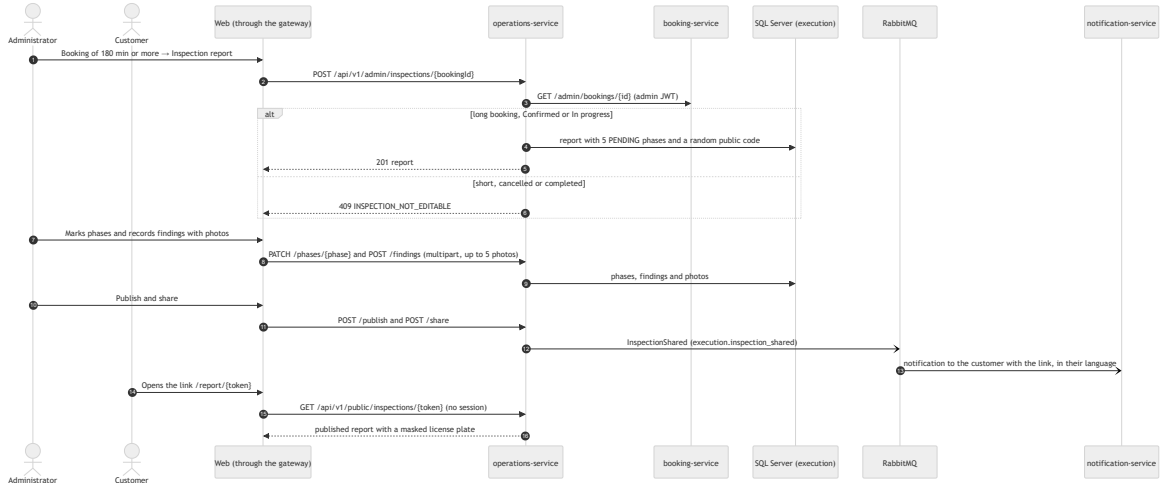
Source: `security-service` (`RolePermissionController`, `RolePermissionAdministrationService`, migration `026-security-role-permissions.sql`), ADR-015.

2.8 Sign-in With Lockout After Failed Attempts (RF-002, RF-004b)



Source: `security-service` (`AuthenticationService.login`, `LoginLockout`, `PasswordRecoveryService`, migration `034-security-login-lockout.sql`).

2.9 Vehicle Inspection Report (RF-036)



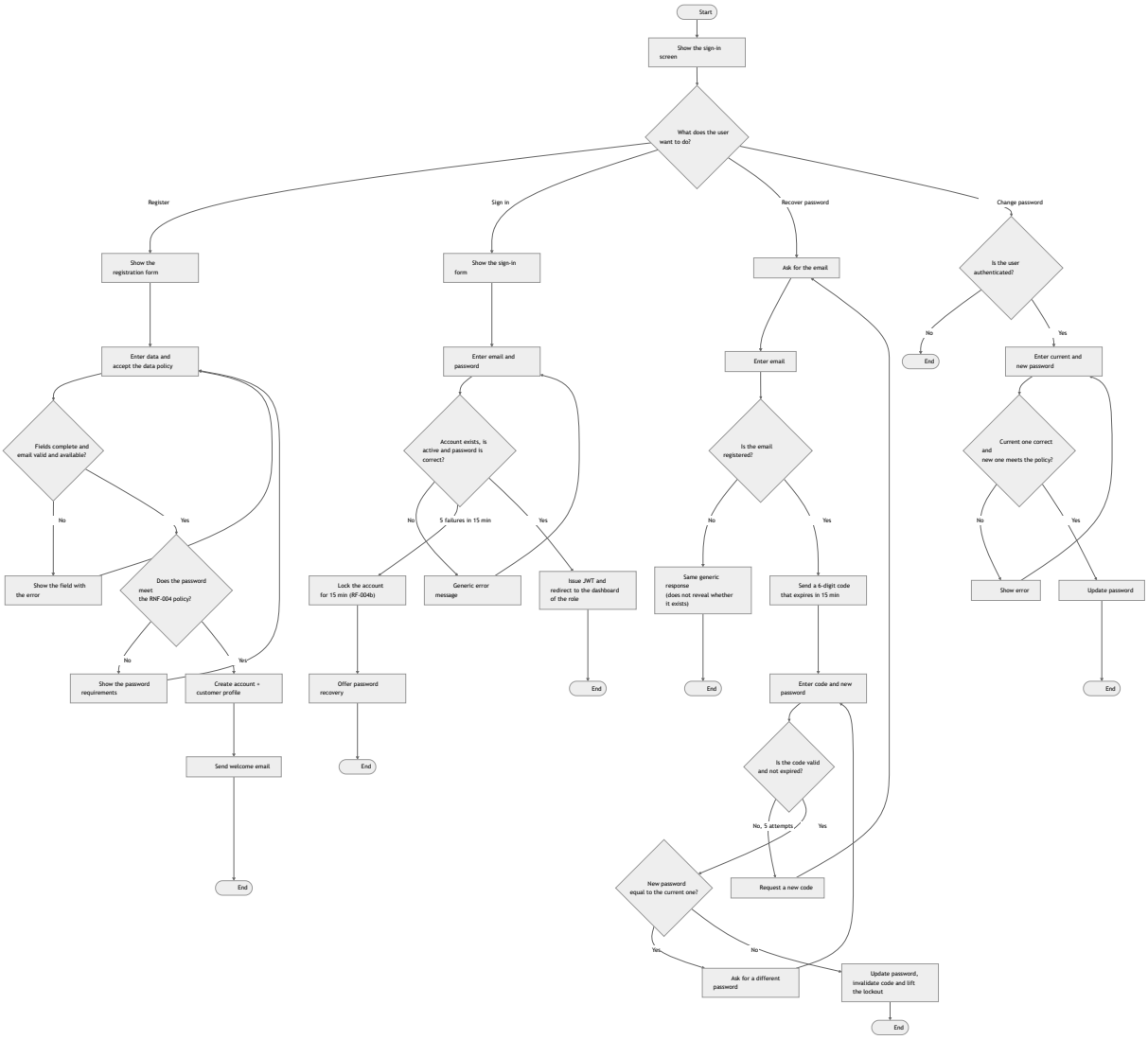
Source: `operations-service` (`InspectionReport`, migration `033-execution-inspection-report.sql`), `notification-service` (shared report notice).

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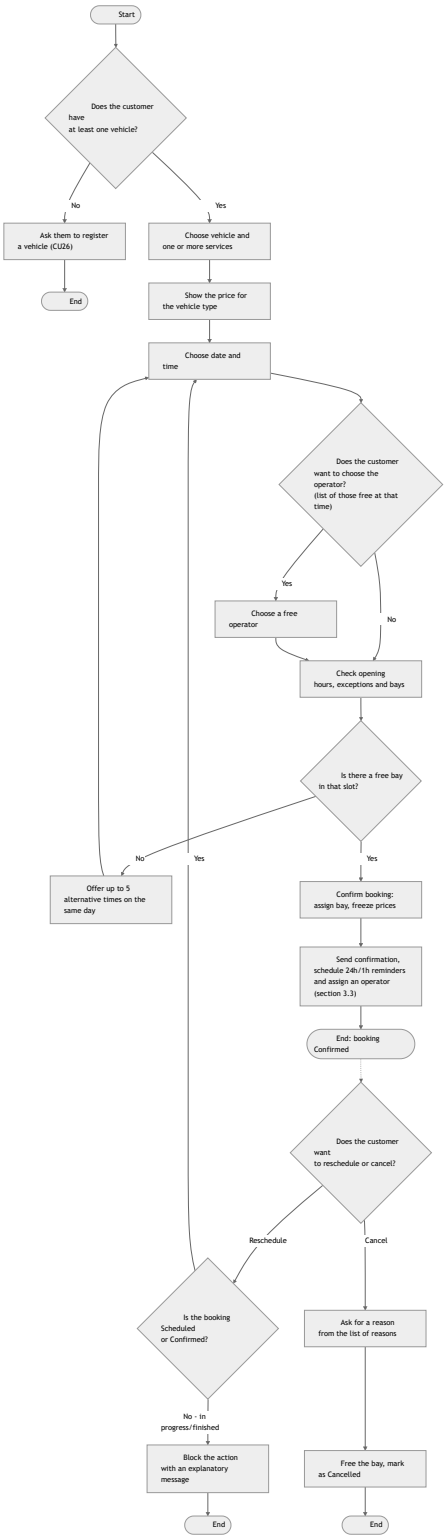
3. Activity Diagrams

The activity diagrams show the business processes as they work in the system. There is no payment gateway: the customer uploads a receipt and the administrator reviews it (ADR-008).

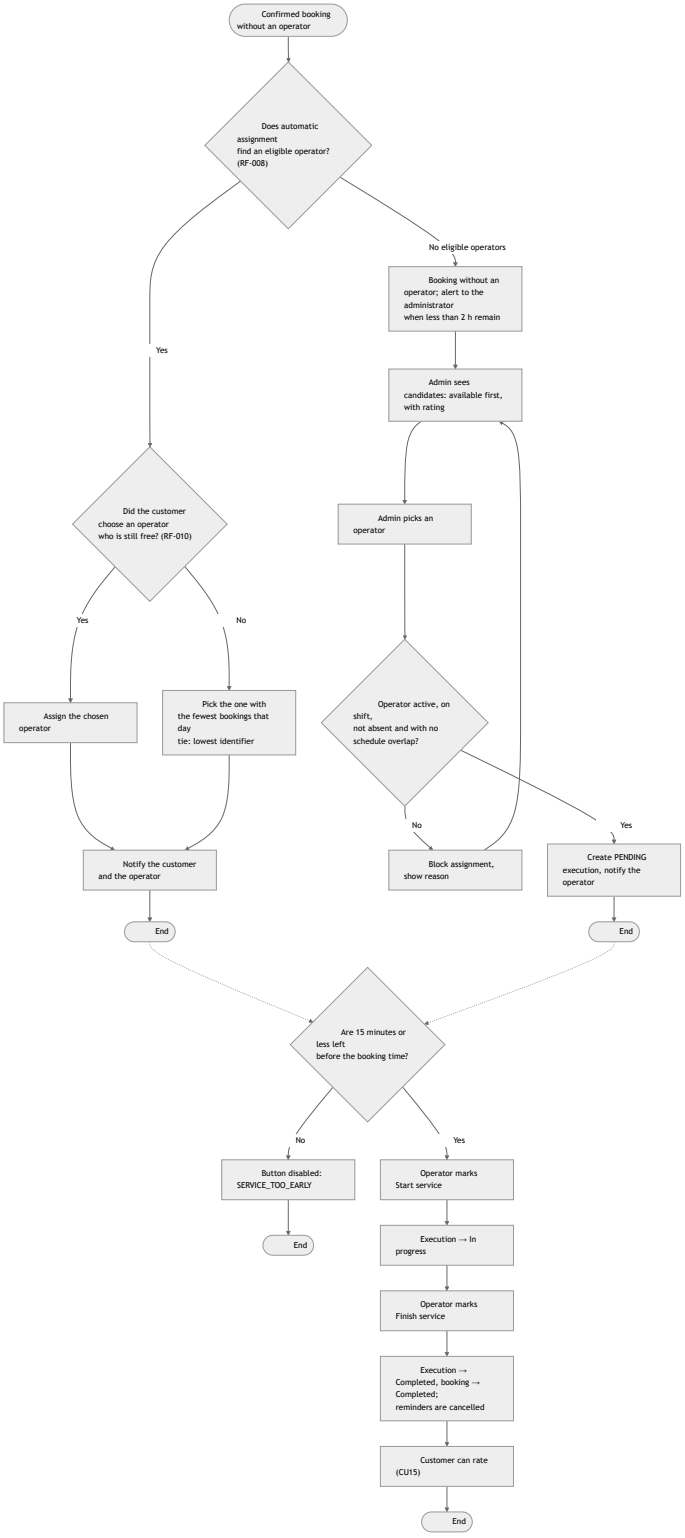
3.1 Authentication (RF-001 to RF-004)



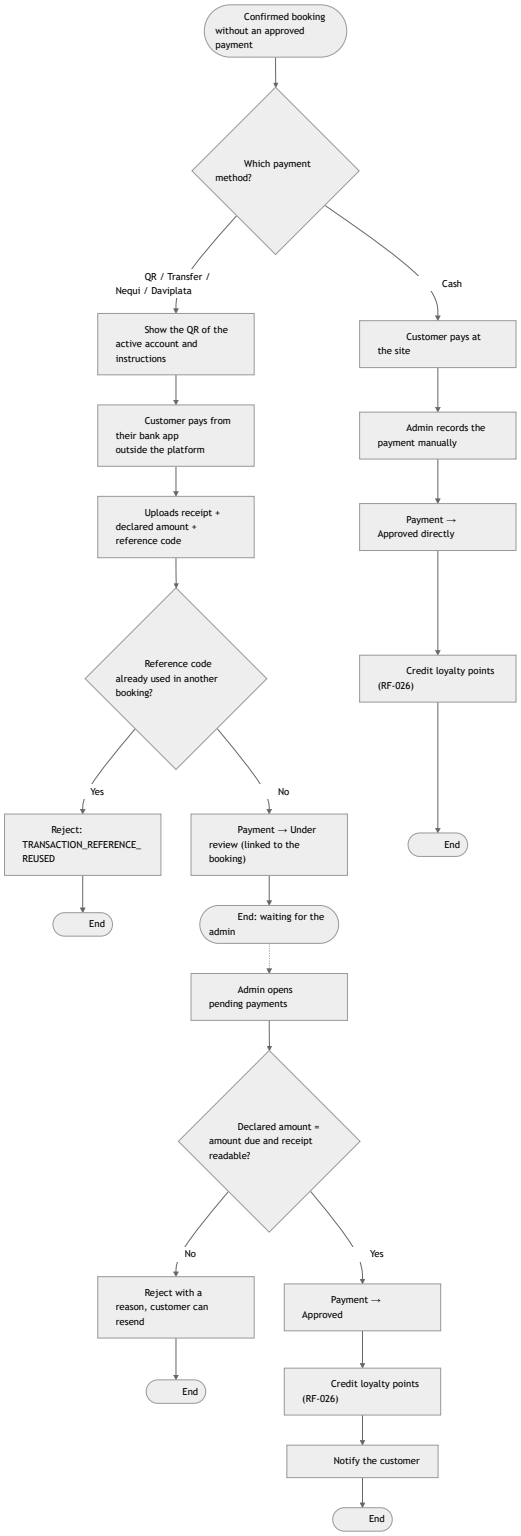
3.2 Booking a Wash (RF-005, RF-006, RF-007, RF-010)



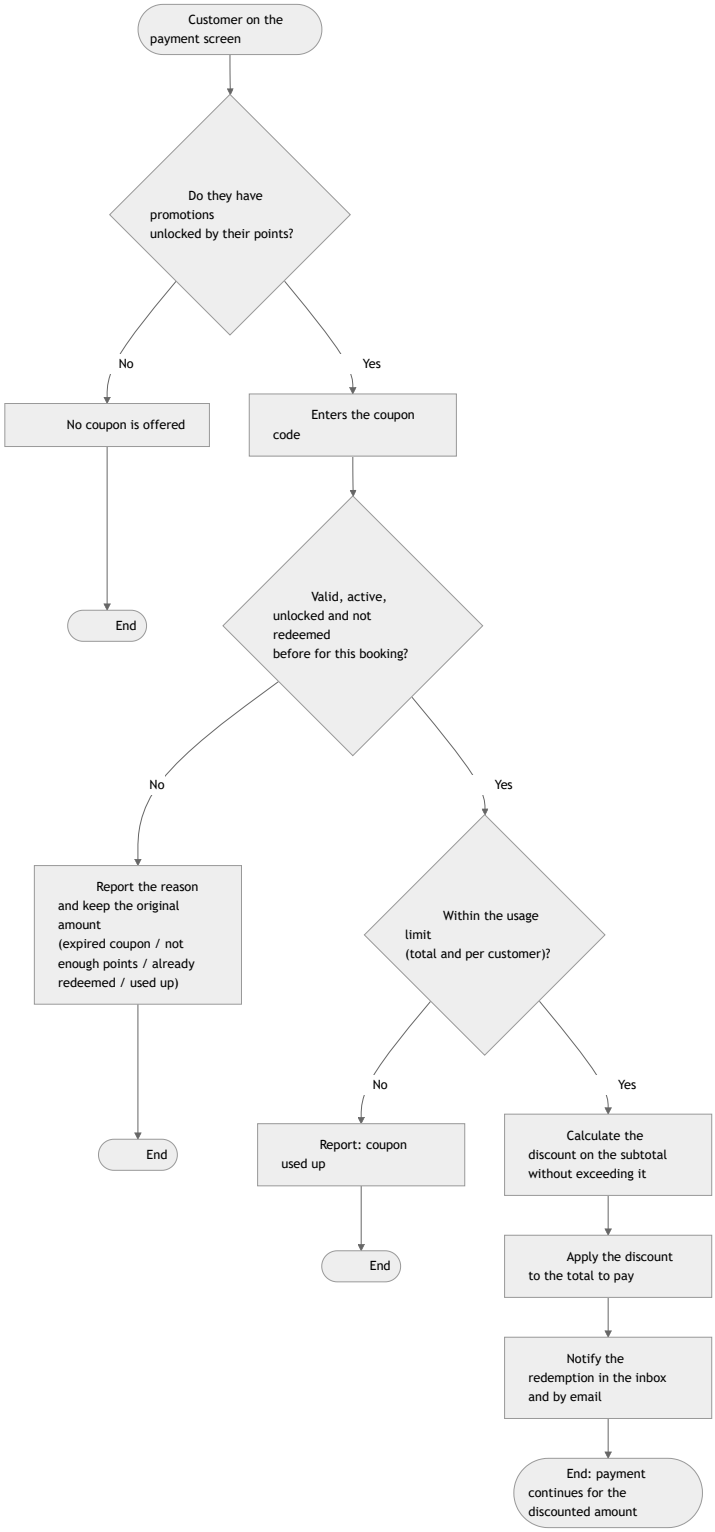
3.3 Operator Assignment and Service Execution (RF-008, RF-009, RF-010, RF-011, RF-012b)



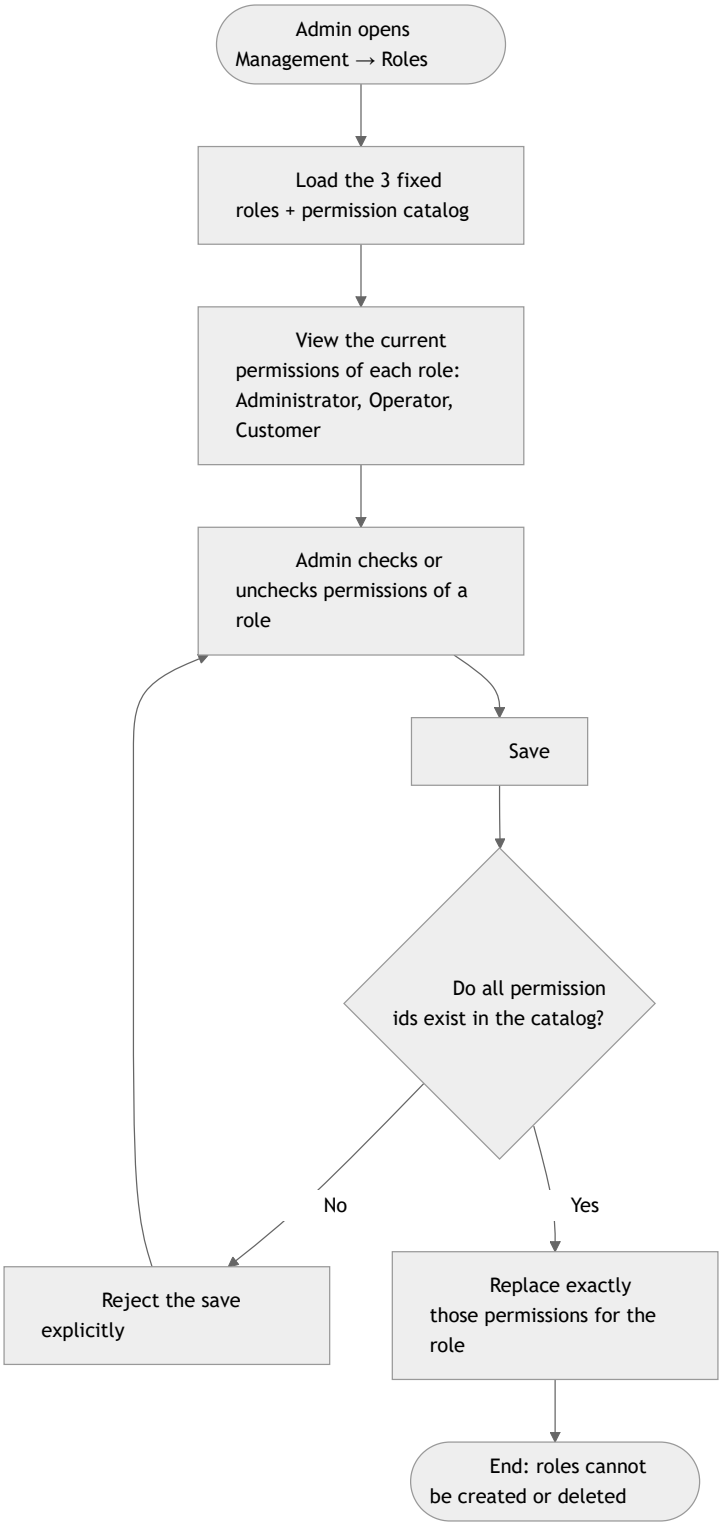
3.4 Paying for a Booking (RF-013, RF-013b, RF-014, RF-016)



3.5 Redeeming a Loyalty Coupon (RF-024, RF-025, RF-026)



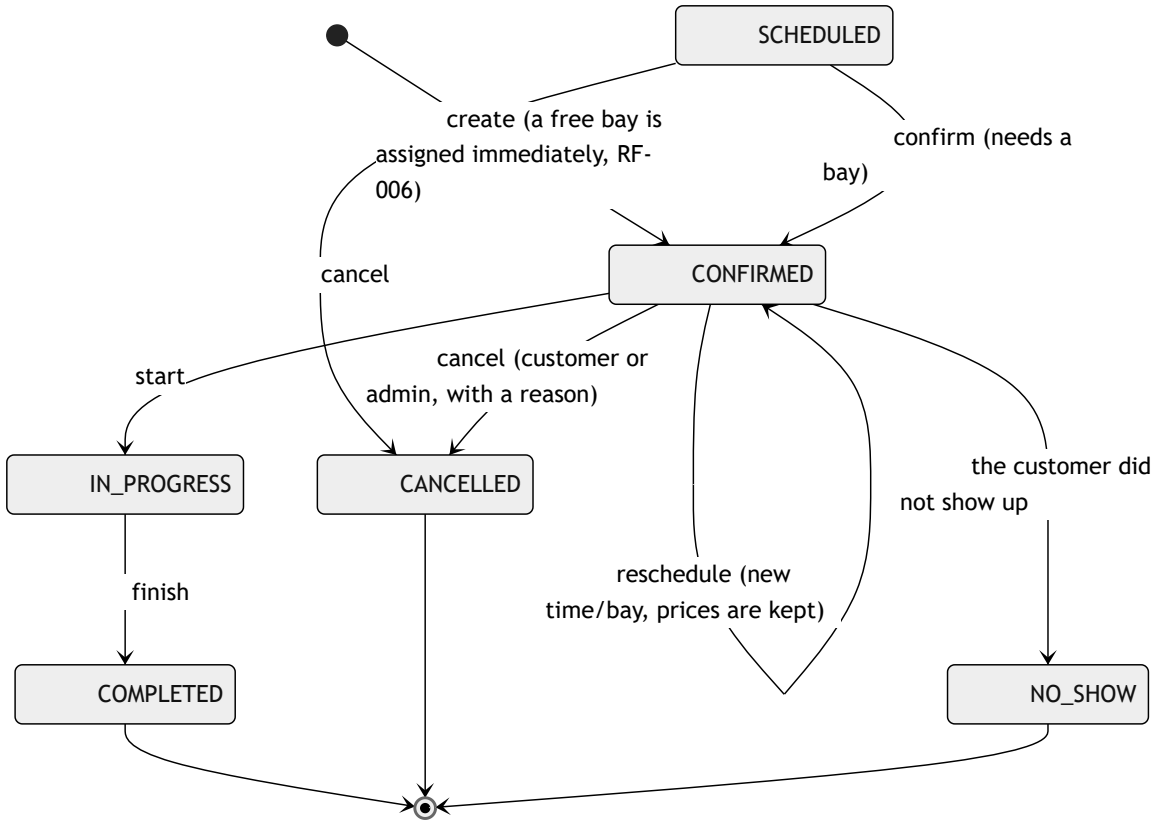
3.6 Managing Roles and Permissions (RF-029)



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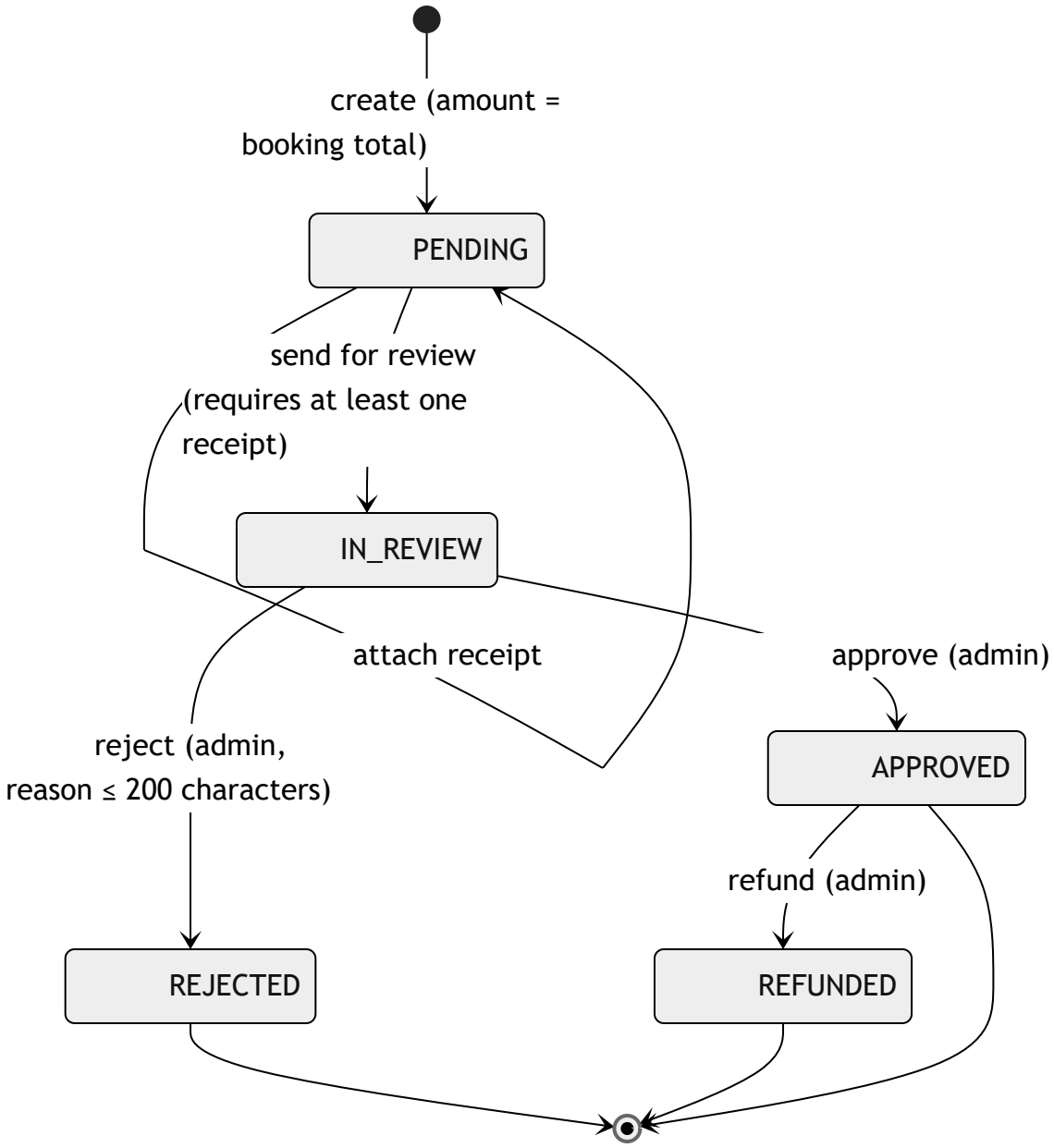
4. State Diagrams

4.1 Booking Life Cycle (RF-006, RF-007)



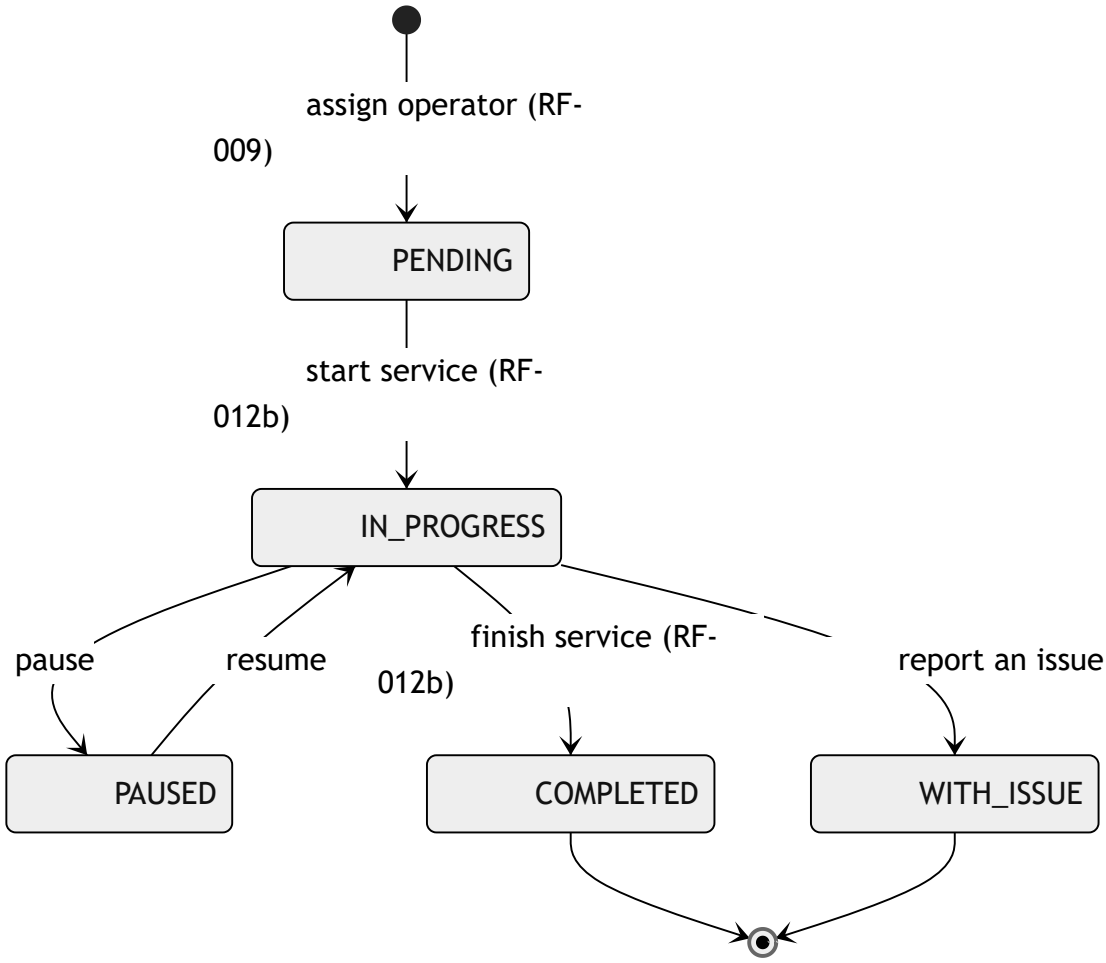
Source: `booking-service`, `Booking.moveTo`, `Booking.cancel`, `BookingStatus`. Only `SCHEDULED` and `CONFIRMED` can be rescheduled or cancelled (RF-007); `SCHEDULED` exists in the model, but the current flow always creates the booking directly as `CONFIRMED`.

4.2 Payment Life Cycle (RF-013b, RF-014, ADR-008)



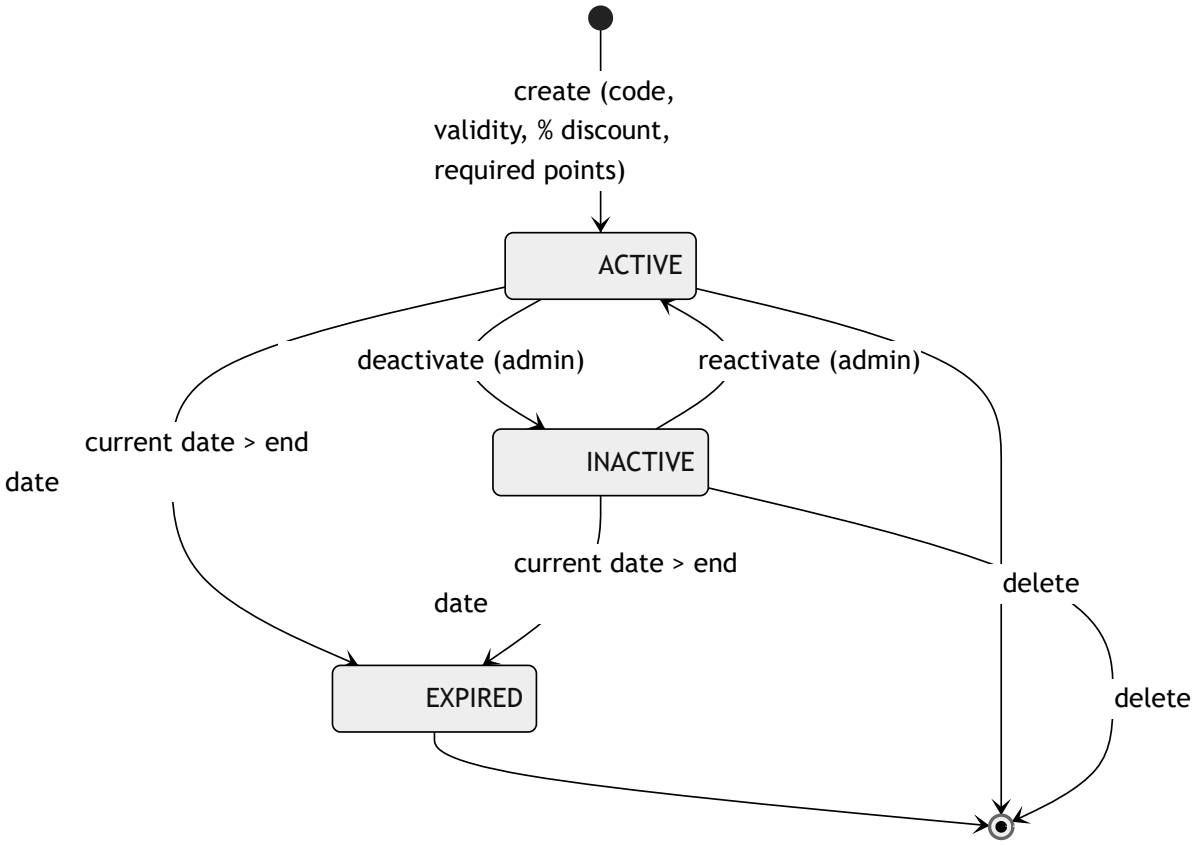
Source: `payment-service`, `Payment` aggregate. A booking allows only one approved payment and only one open payment at a time; after `REJECTED` the customer can report a new payment.

4.3 Service Execution Life Cycle (RF-012b)



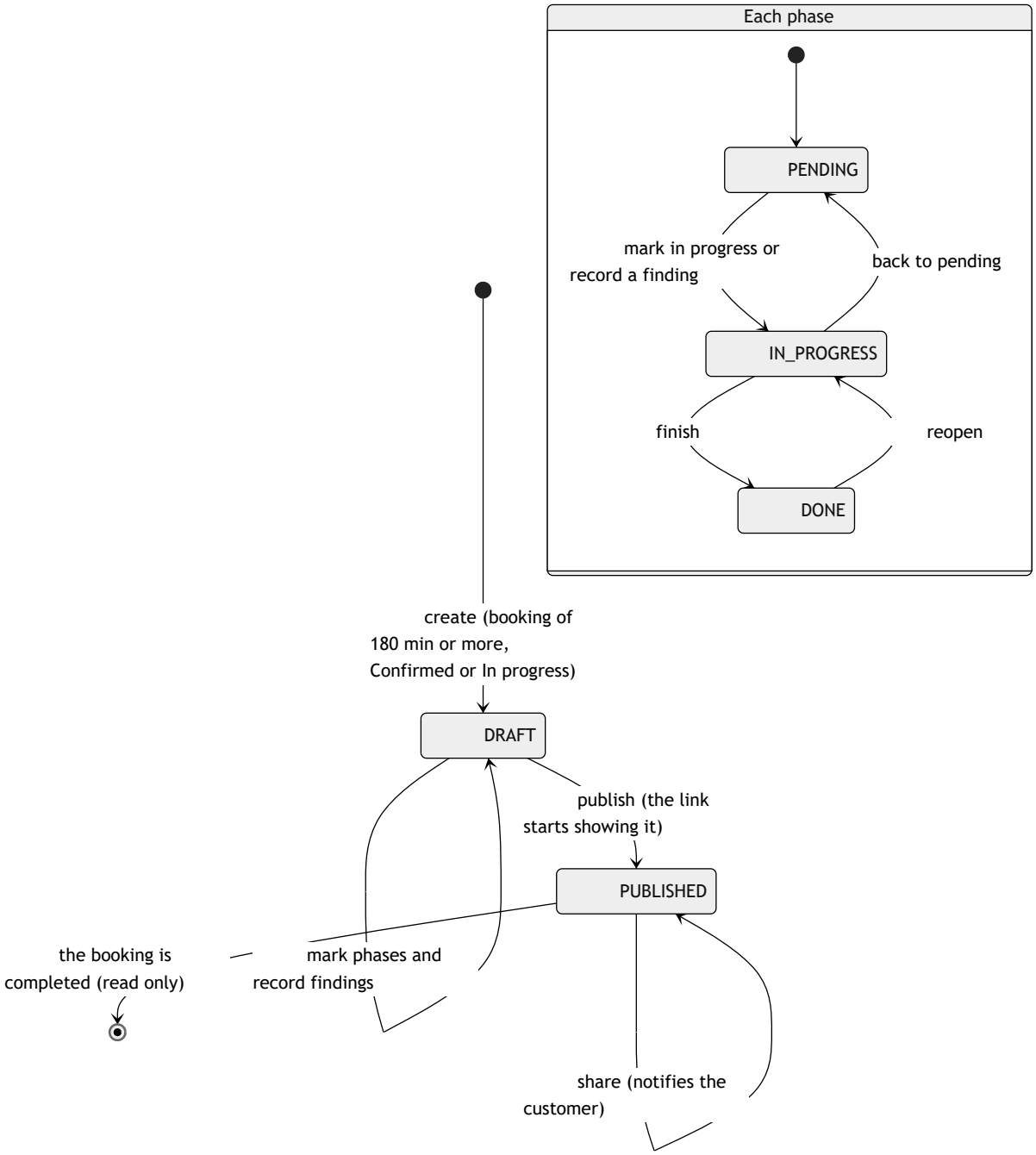
Source: `operations-service`. `COMPLETED` enables the customer's rating (RF-012).

4.4 Promotion Life Cycle (RF-024)



Source: `payment-service`, `Promotion.IsAvailableOn` (active and within the validity period). An `INACTIVE` or `EXPIRED` promotion cannot be redeemed (RF-024, RN-PROMO-01).

4.5 Inspection Report and Its Phases (RF-036)



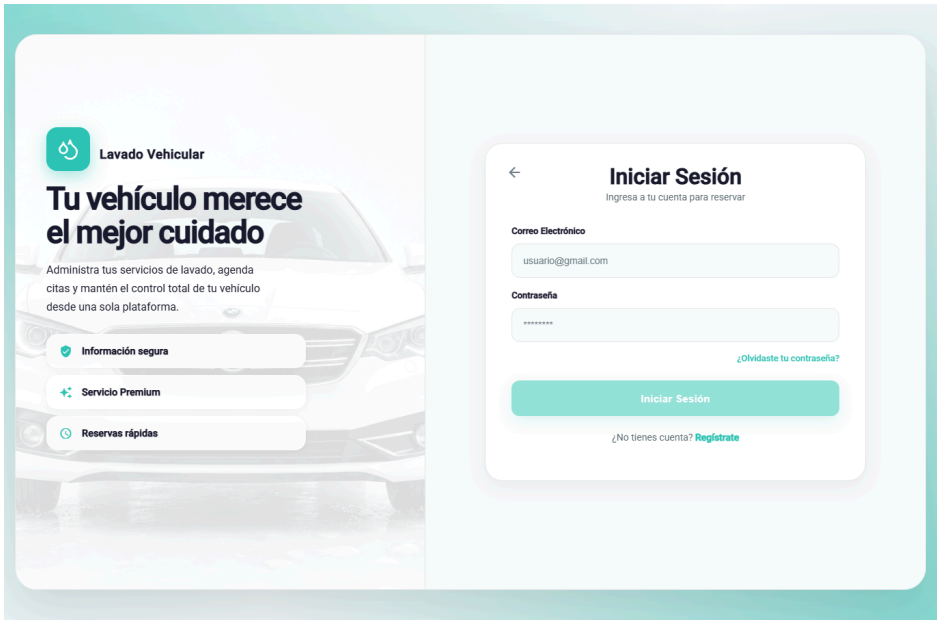
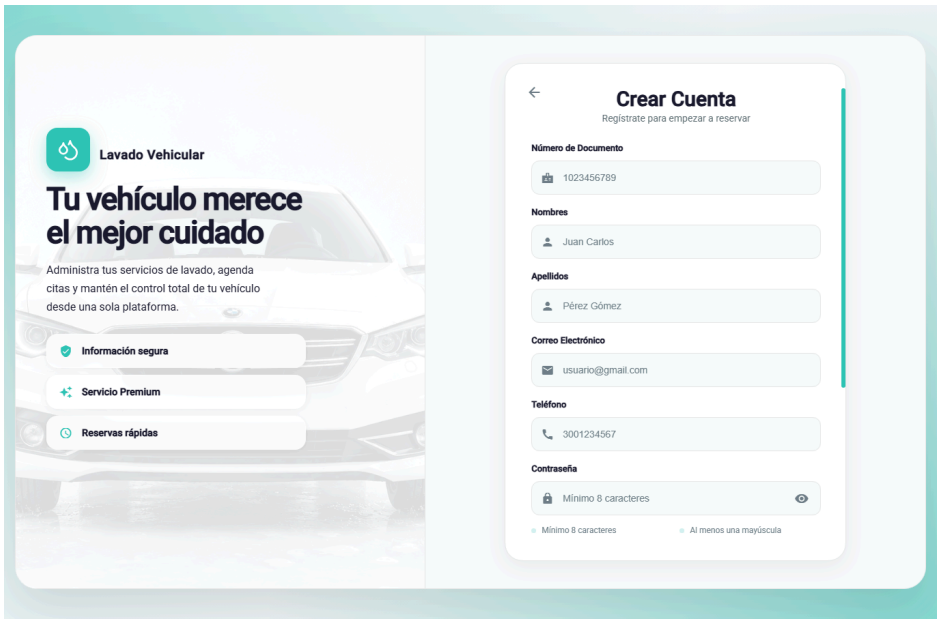
Source: `operations-service`, `InspectionReport` (`publish`, `requirePublished`, `setPhaseStatus`, `addFinding`). The phases are reception, exterior, interior, engine and delivery; only a published report can be shared.

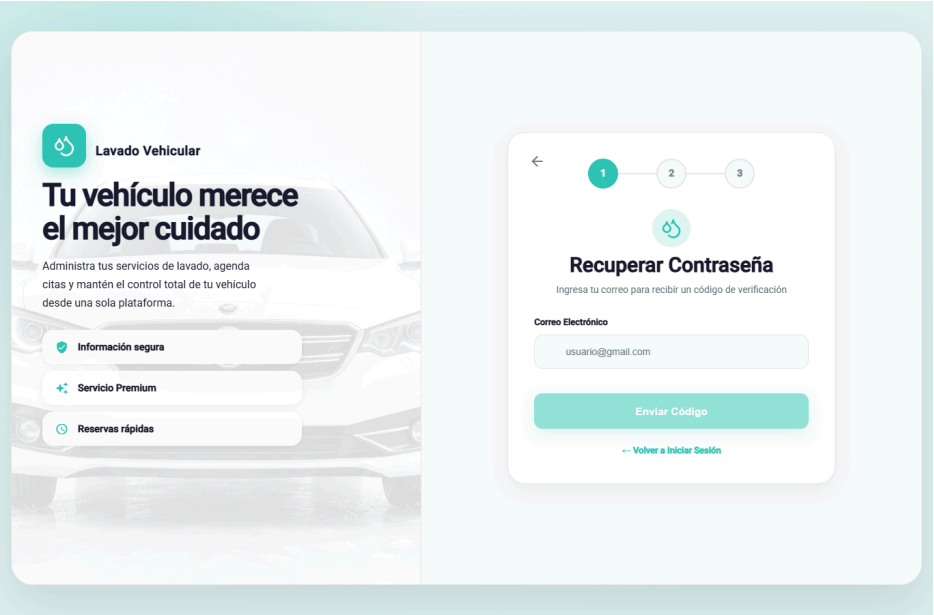
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5. Screen Inventory (Mockup)

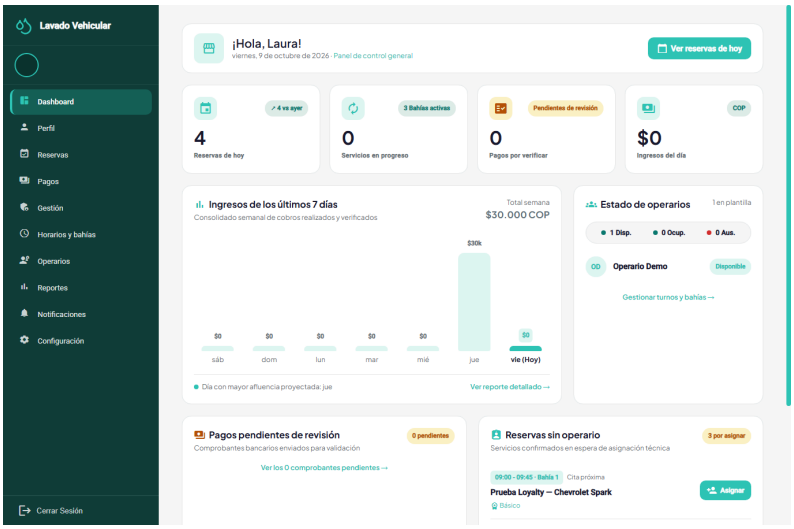
Screen by screen and by role, including the create and edit modals of each module. All screenshots are in the `capturas/` folder of this document.

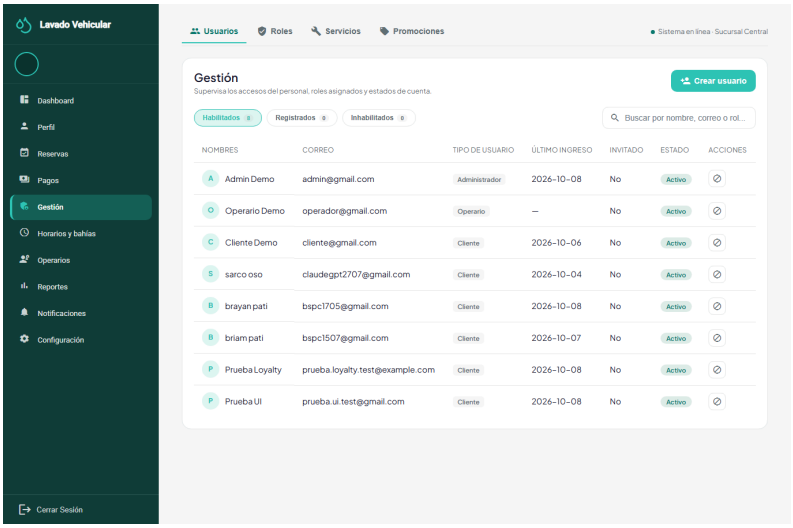
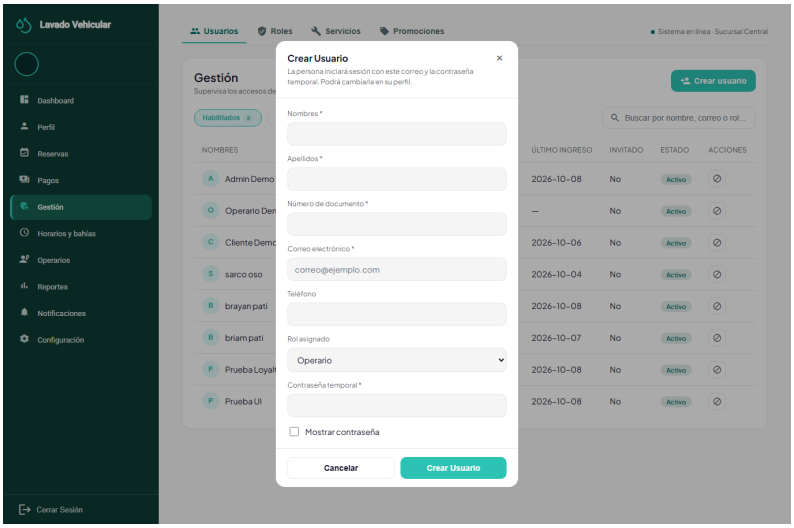
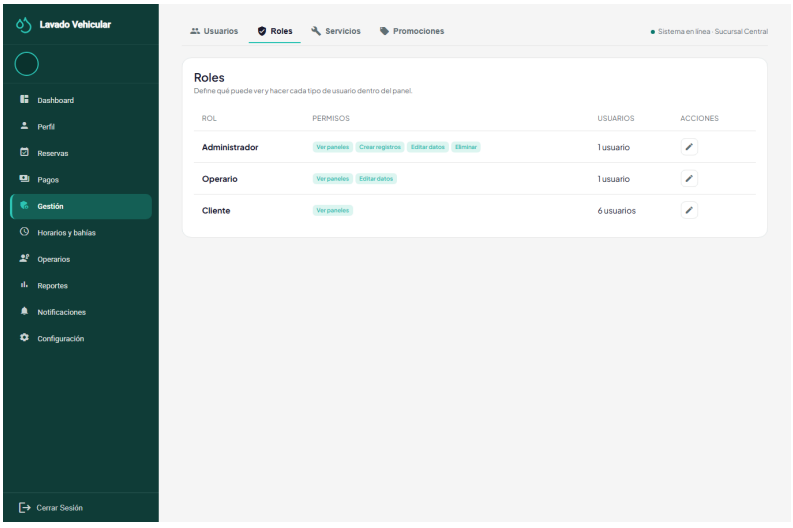
5.1 Common Screens (No Session)

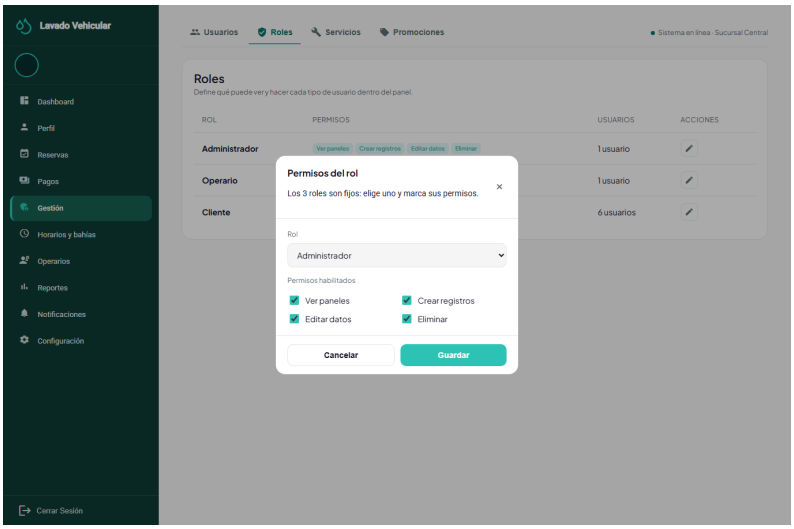
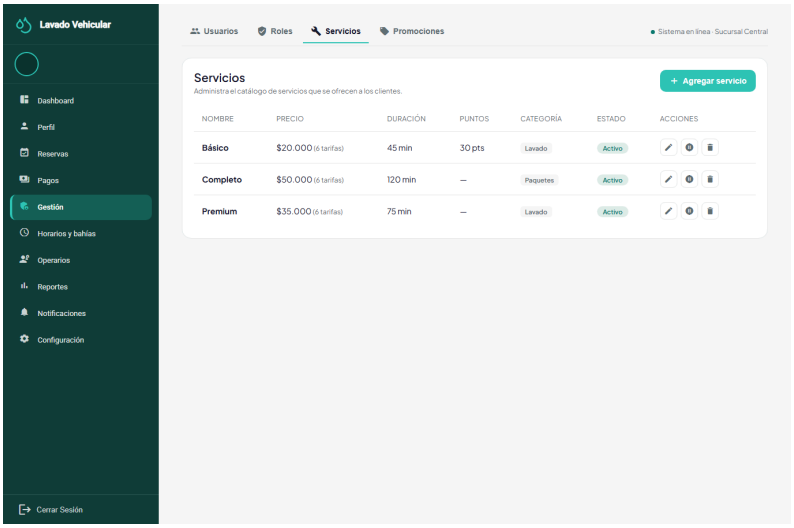
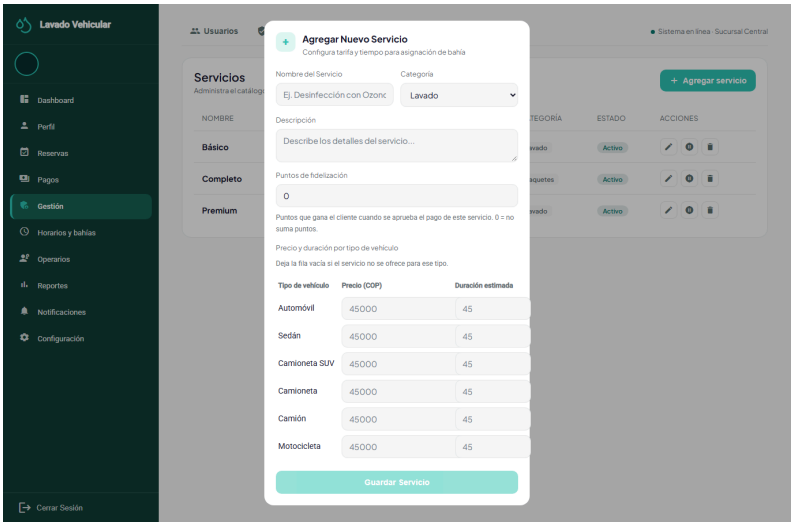
Screen	Evidence
Sign-in	
Registration	

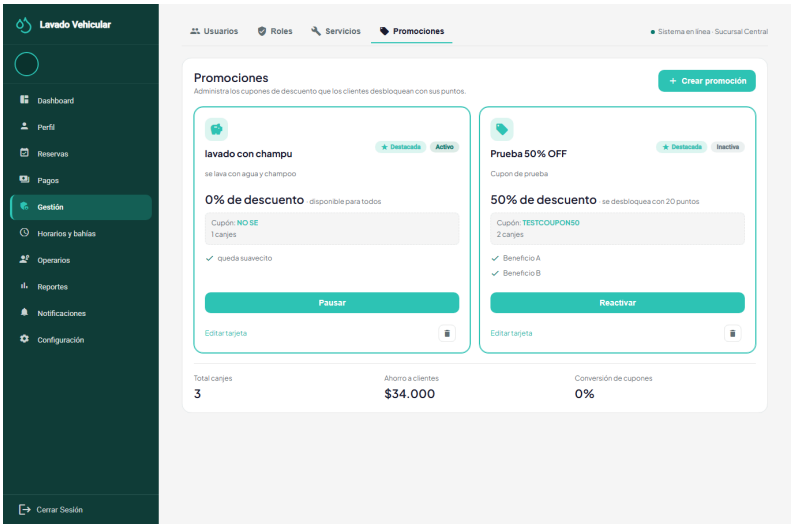
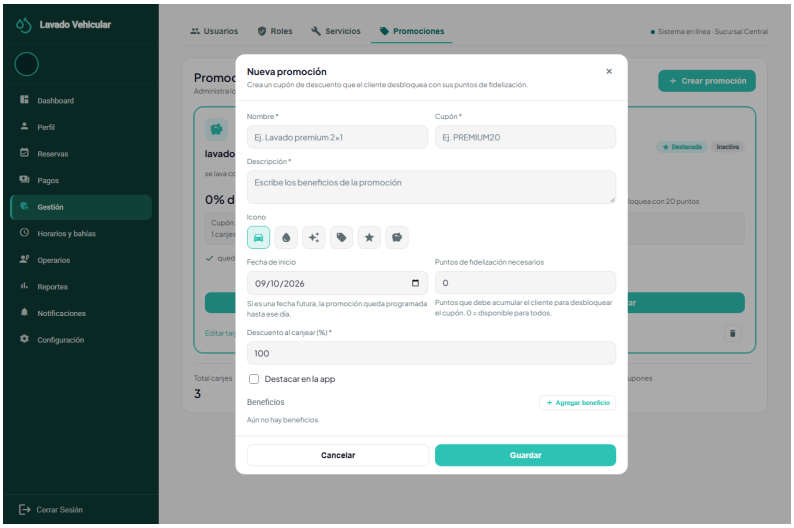
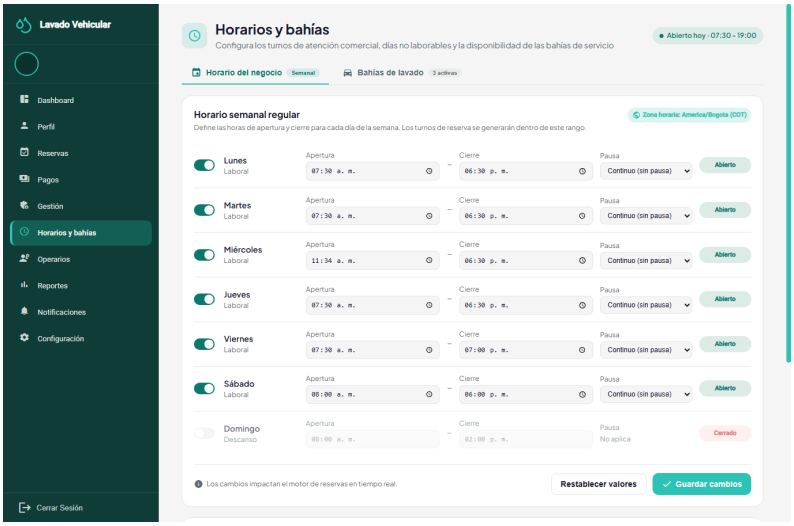
Screen	Evidence
Password recovery	

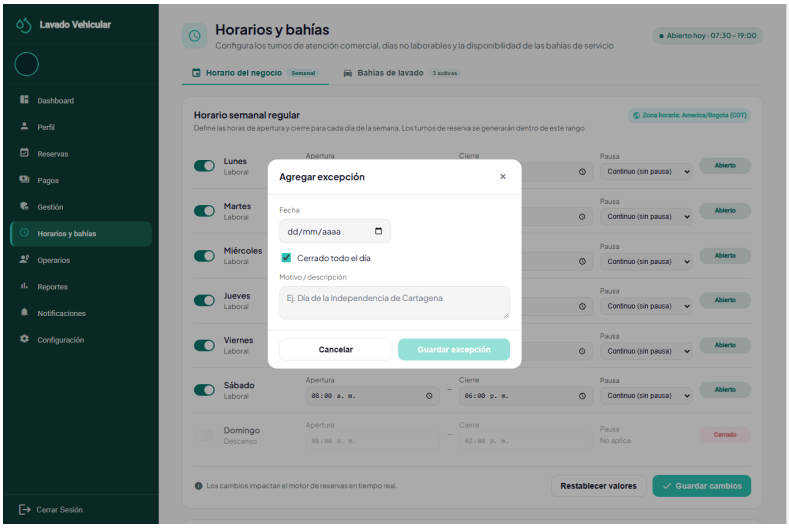
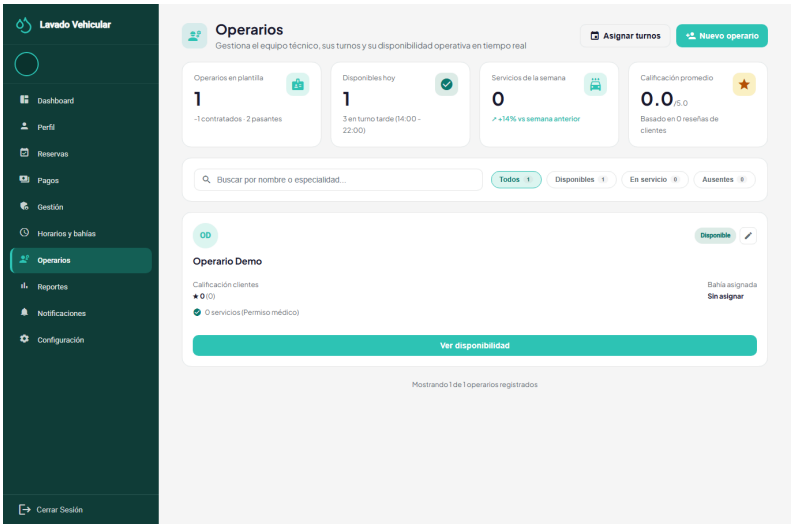
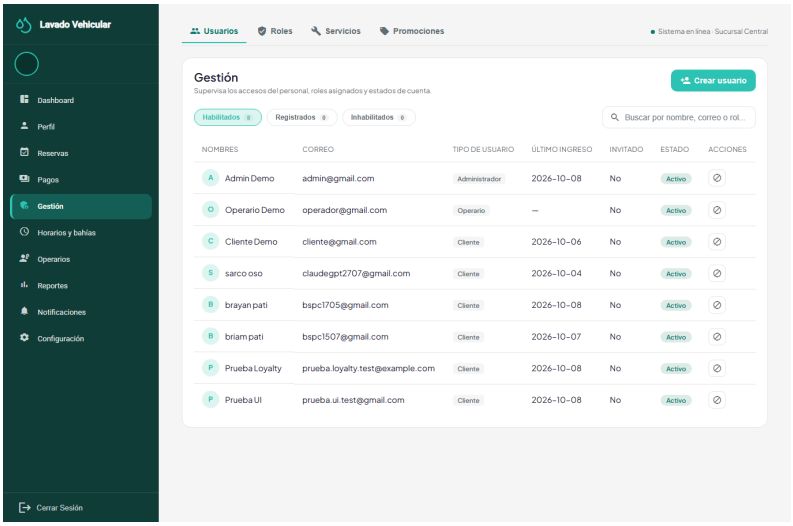
5.2 Administrator Role

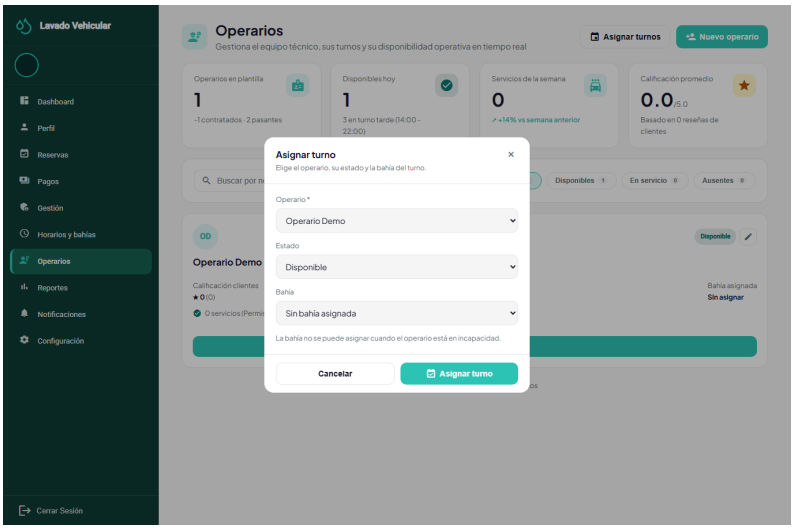
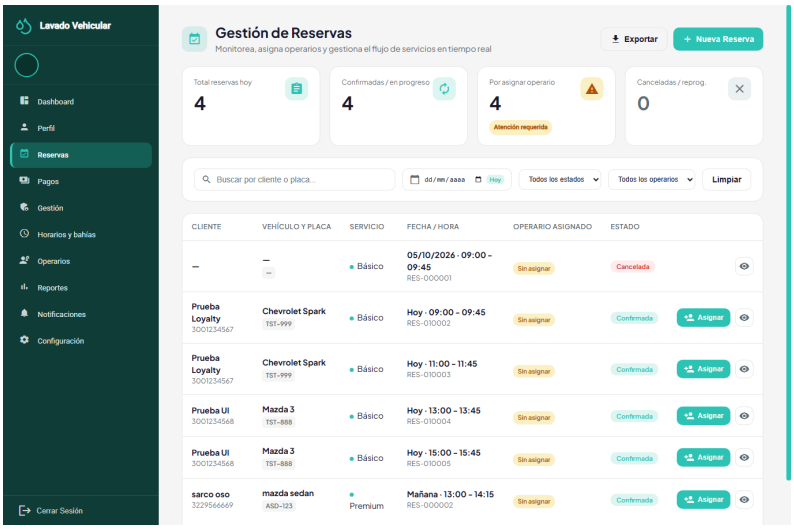
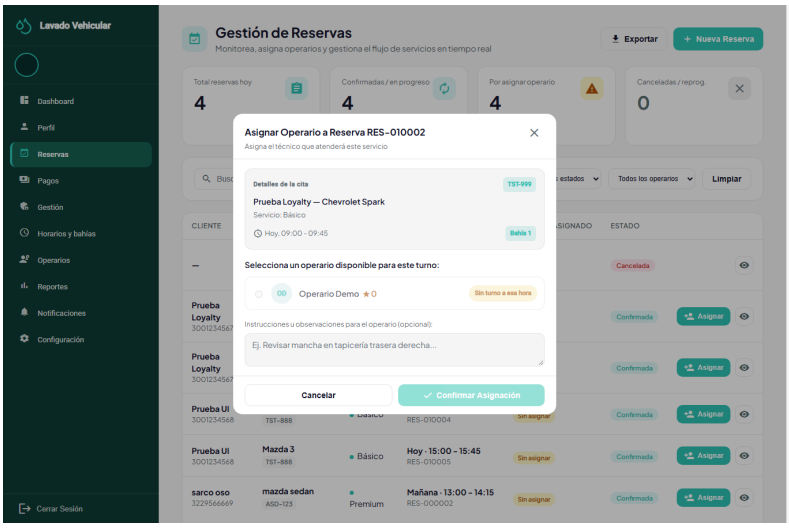
Screen	Evidence
Dashboard	

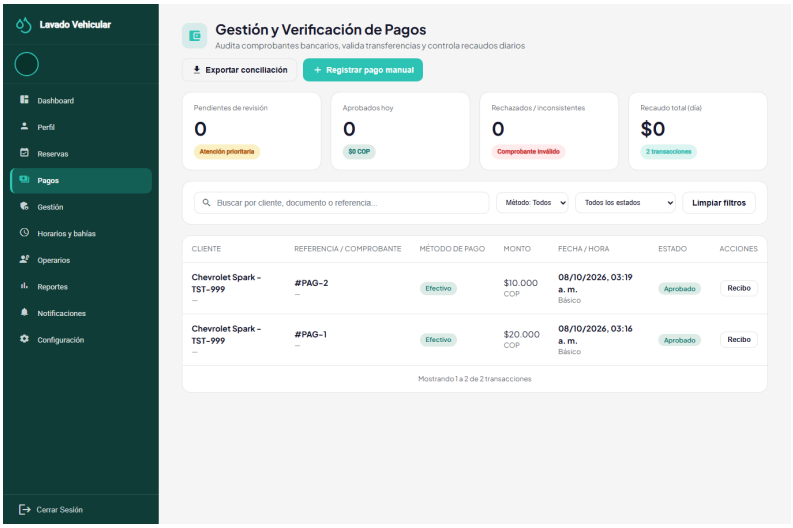
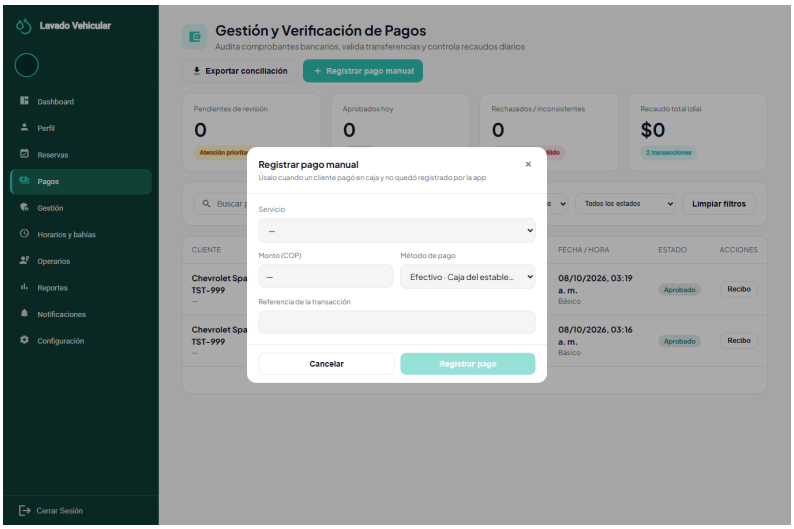
Screen	Evidence
Management → Users	
Modal: Create user	
Management → Roles	

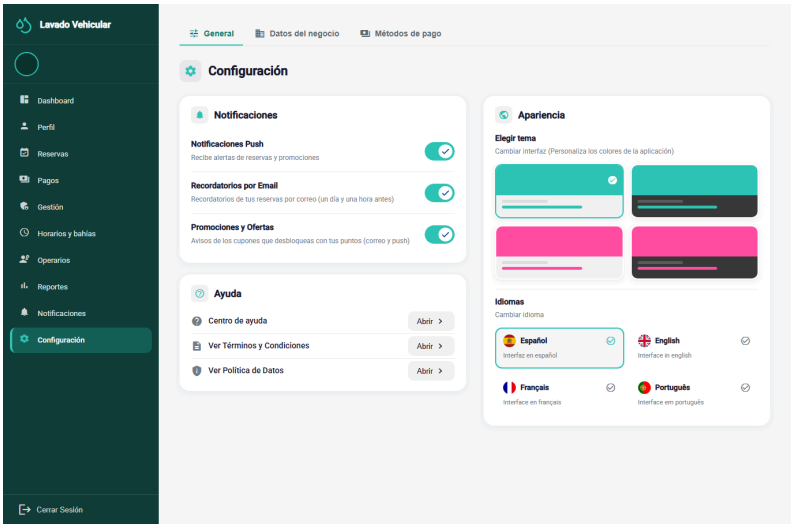
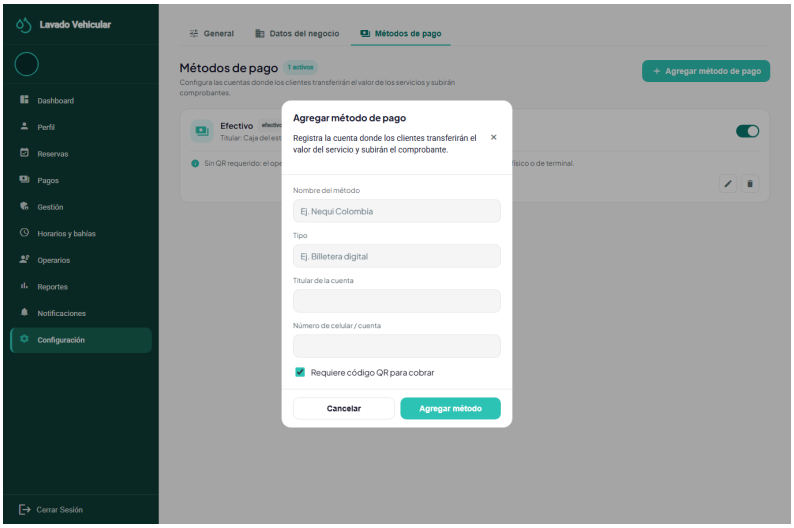
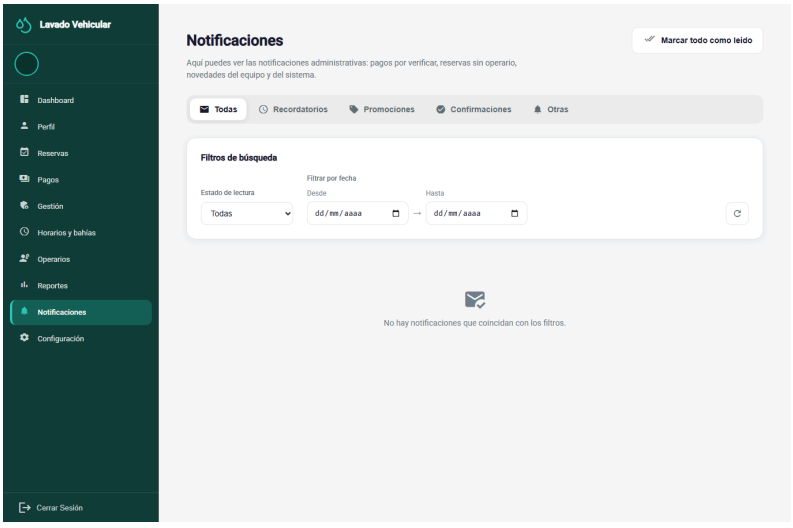
Screen	Evidence
Modal: Edit role permissions	
Management → Services (catalog)	
Modal: Add service	

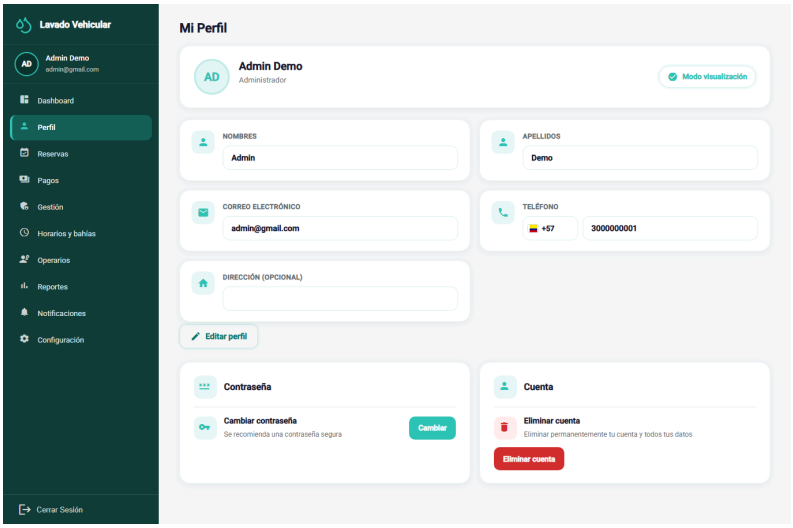
Screen	Evidence
Management → Promotions	
Modal: Create promotion	
Opening hours and bays	

Screen	Evidence
Modal: Add opening-hours exception	
Operators	
Modal: New operator	

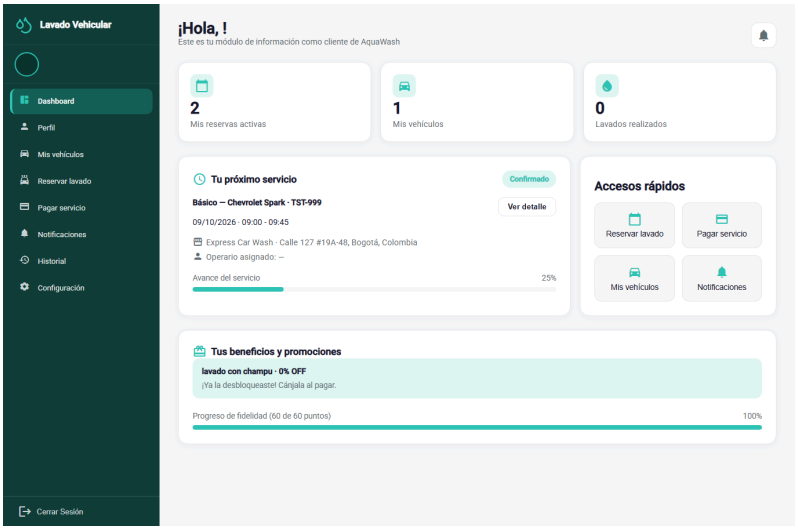
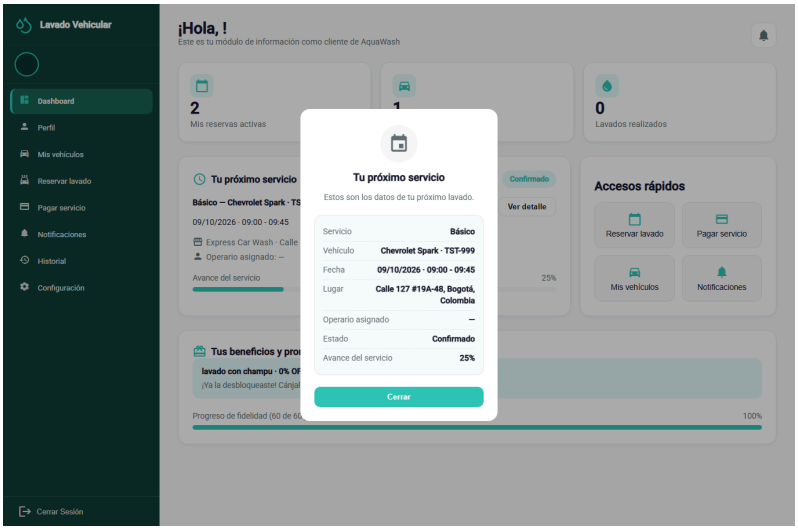
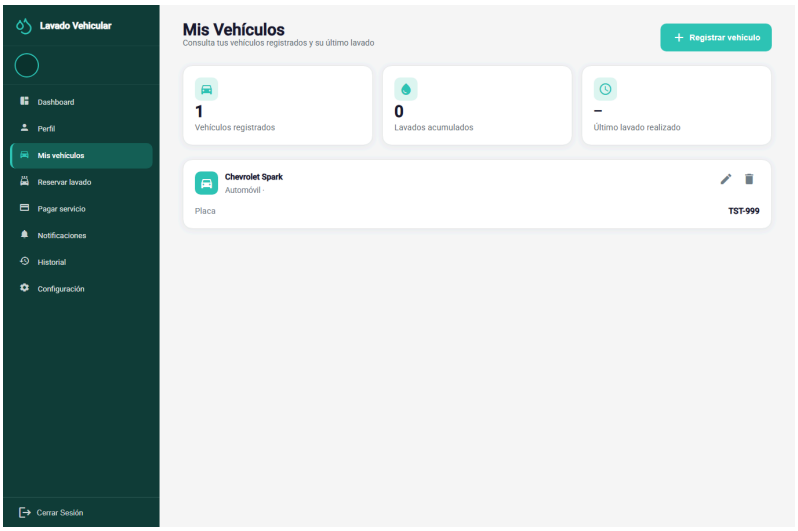
Screen	Evidence
Modal: Assign shift	
Bookings	
Modal: Assign an operator to a booking	

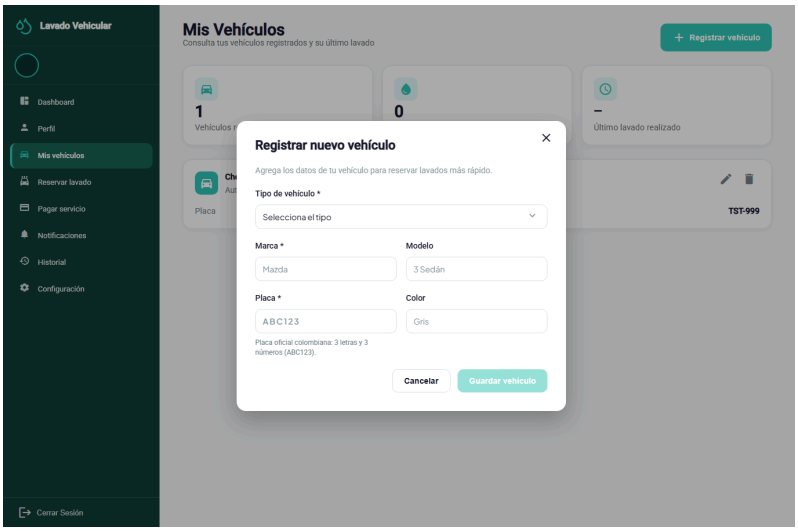
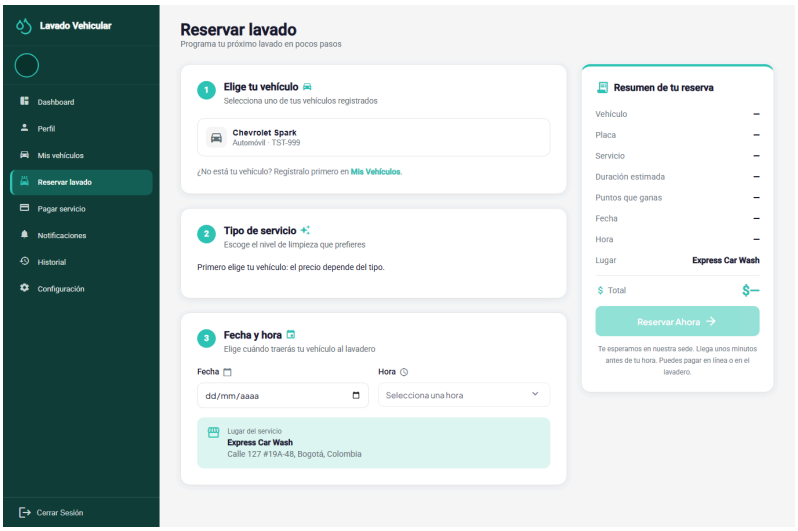
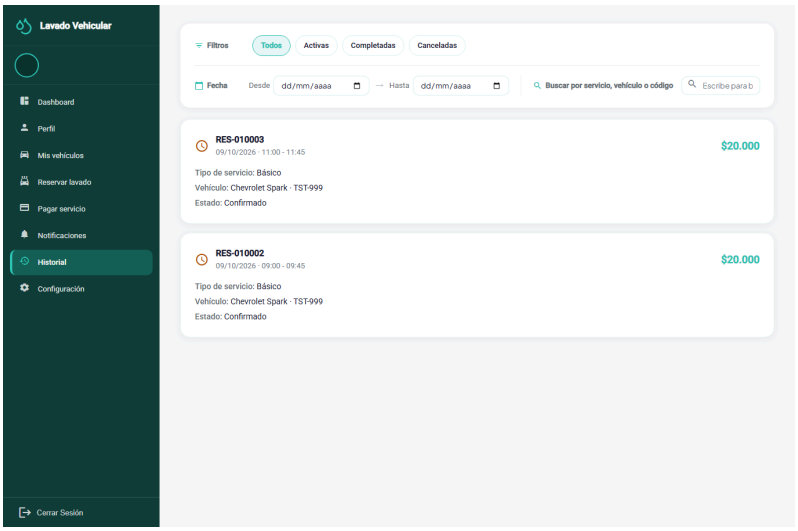
Screen	Evidence
Payments	
Modal: Record manual payment	
Reports	

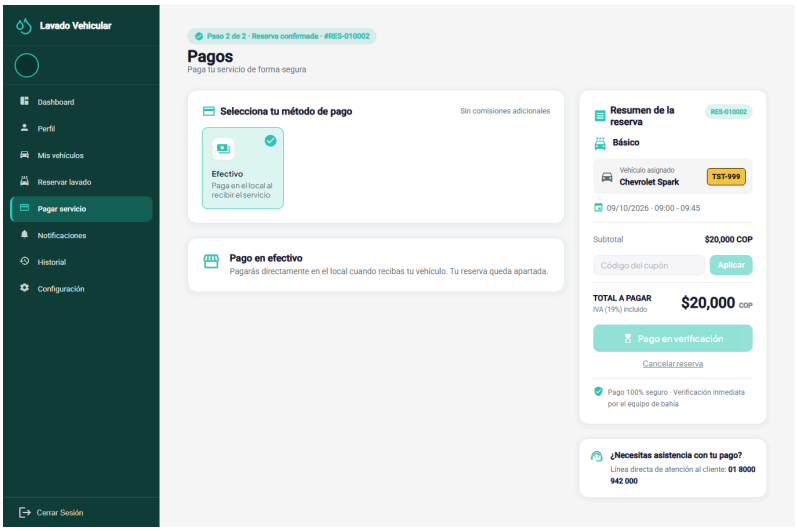
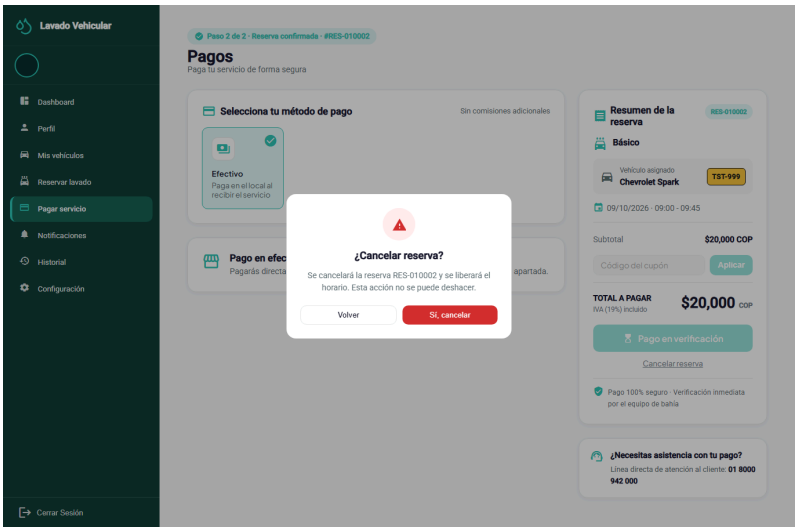
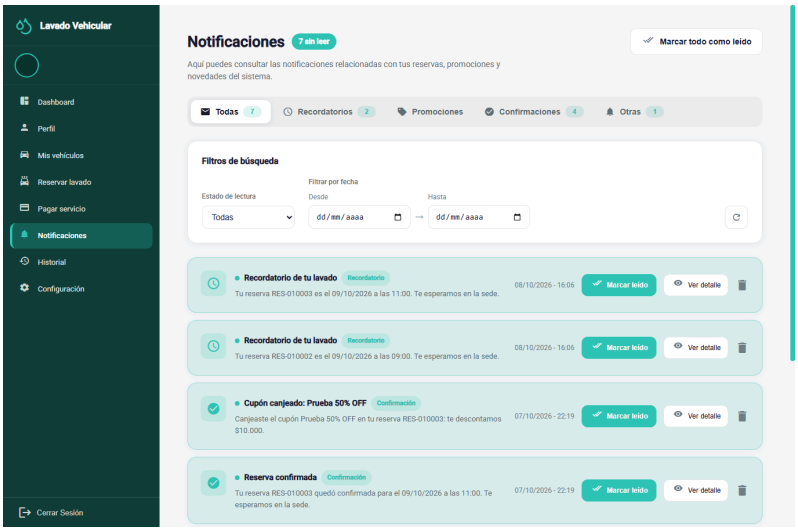
Screen	Evidence
Settings → Payment accounts	
Modal: Add payment account	
Notifications	

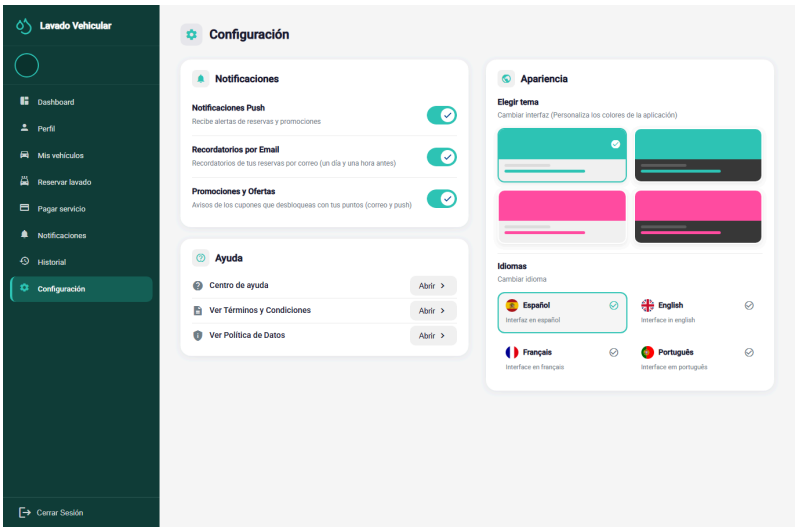
Screen	Evidence
Profile	

5.3 Customer Role

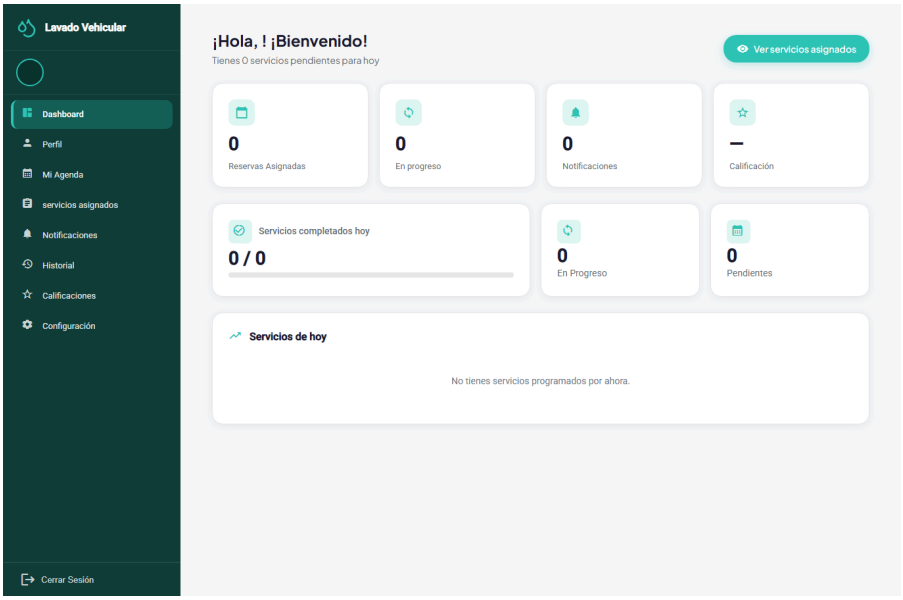
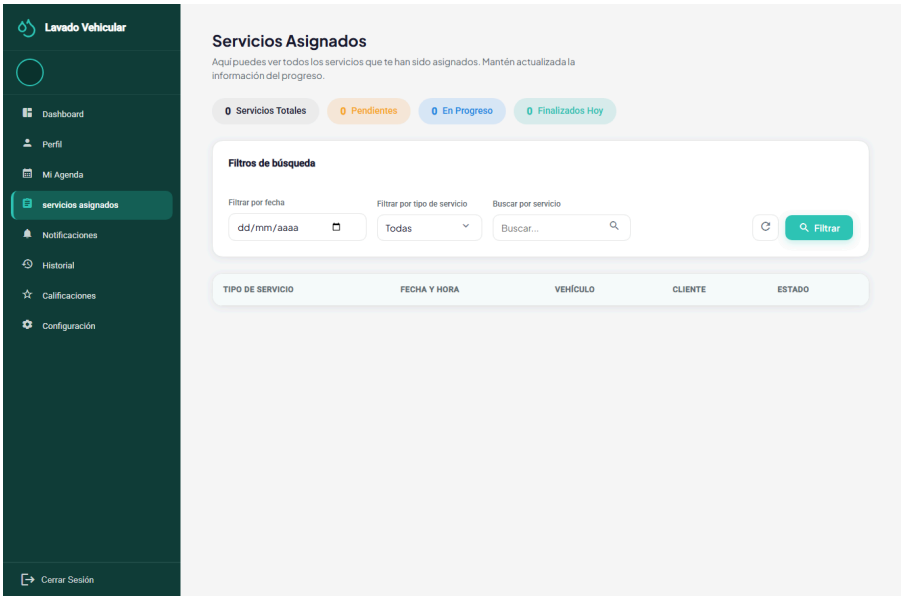
Screen	Evidence
Dashboard	
Modal: View details of the next service	
My vehicles	

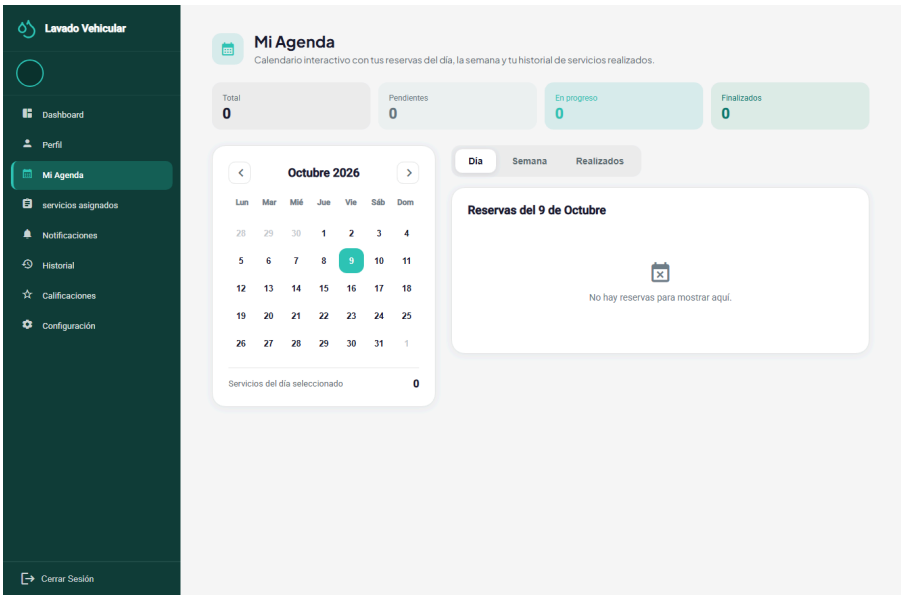
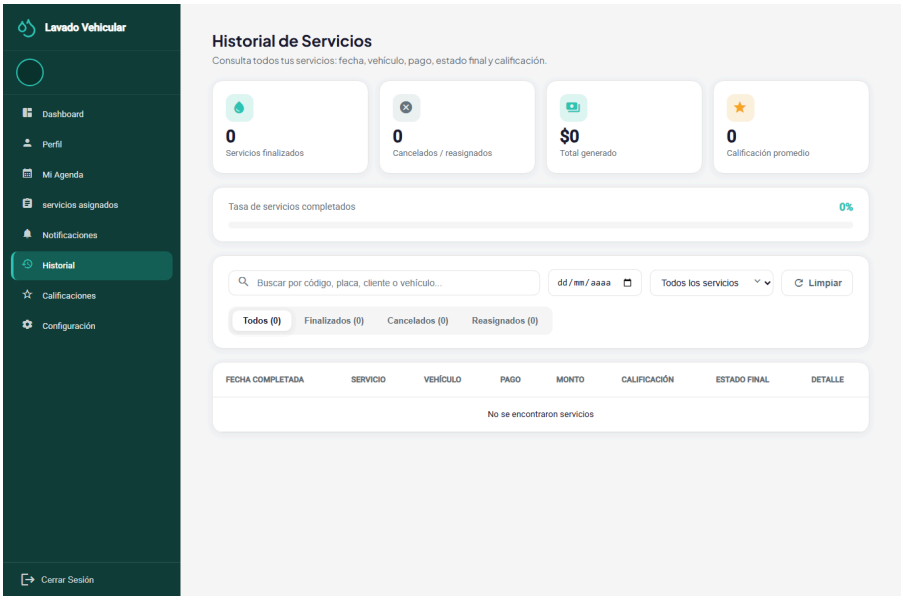
Screen	Evidence
Modal: Register vehicle	
Book a wash	
Booking history	

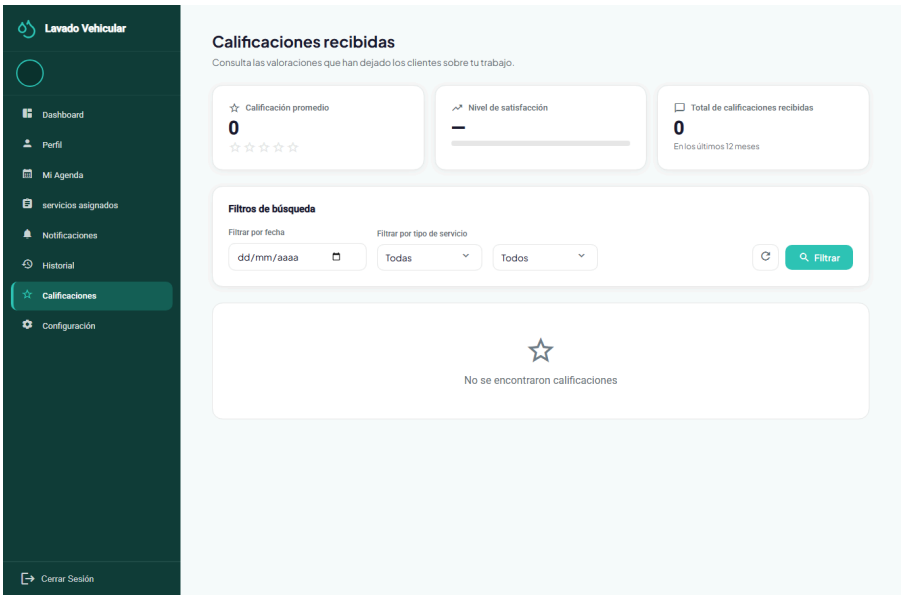
Screen	Evidence
Pay for a service	
Modal: Confirm booking cancellation	
Notifications	

Screen	Evidence
Settings (theme and language)	

5.4 Operator Role

Screen	Evidence
Main dashboard	
Assigned services	

Screen	Evidence
My schedule (availability)	
Service history	

Screen	Evidence
Ratings received	

Correlation With the Software Requirements Specification

Dynamic view in this document	Requirements it illustrates
§2.1 Registration and welcome	RF-001, RF-018
§2.2 Booking a wash	RF-005, RF-006, RF-010
§2.3 Reminders and alert	RF-018
§2.4 Assignment and execution	RF-008, RF-009, RF-010, RF-011, RF-012, RF-012b
§2.5 Payment by receipt	RF-013, RF-013b, RF-014, RF-016
§2.6 Coupon redemption and points	RF-024, RF-025, RF-026
§2.7 Roles and permissions	RF-029
§2.8 Sign-in with lockout	RF-002, RF-003, RF-004b
§2.9 Inspection report	RF-036, RF-018
§3.1 to §3.6 Activities	The whole authentication, booking, operations, payment and promotion module
§4.1 to §4.5 States	RF-006, RF-007, RF-012b, RF-013b, RF-014, RF-024, RF-036